Principles of Anatomy and Physiology

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Sixteenth Edition

Chapter 3

The Cellular Level of Organization

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Introduction

- The purpose of this chapter is to:
  - Describe the major components of the cell
  - Discuss the structure and function of the plasma membrane and its proteins
  - Outline the steps of transcription and translation
  - Compare and contrast mitosis and meiosis; explains the event of each process
  - Understand the effects aging has on the cell

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Parts of a Cell

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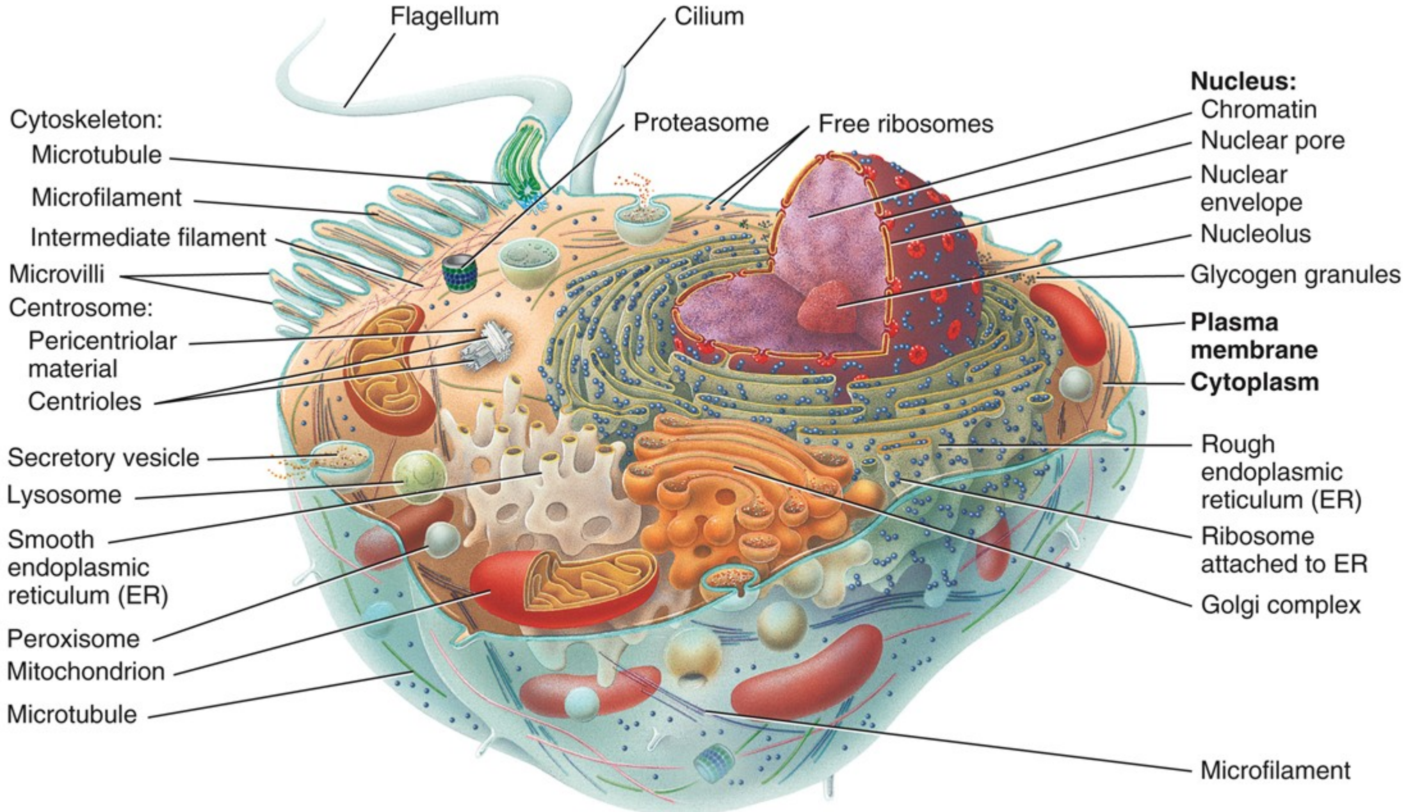
Structures of a Cell



Sectional view

[Cell Anatomy Link]

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Parts of a Cell

- The cell can be subdivided into 3 parts:
  - Plasma (cell) membrane, or plasmalemma
    - Flexible outer boundary
  - Cytoplasm
    - Cytosol – intracellular fluid
    - Organelles – structures that perform a particular function
  - Nucleus
    - Chromosomes
      - DNA and associated proteins

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The Plasma Membrane

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The Plasma Membrane

- Flexible, but sturdy, barrier that surrounds and contains the cytoplasm of the cell

- Fluid mosaic model
  - "mosaic" due to it being composed of many small "pieces"
  - "fluid" since components (like phospholipids) aren't fixed in position

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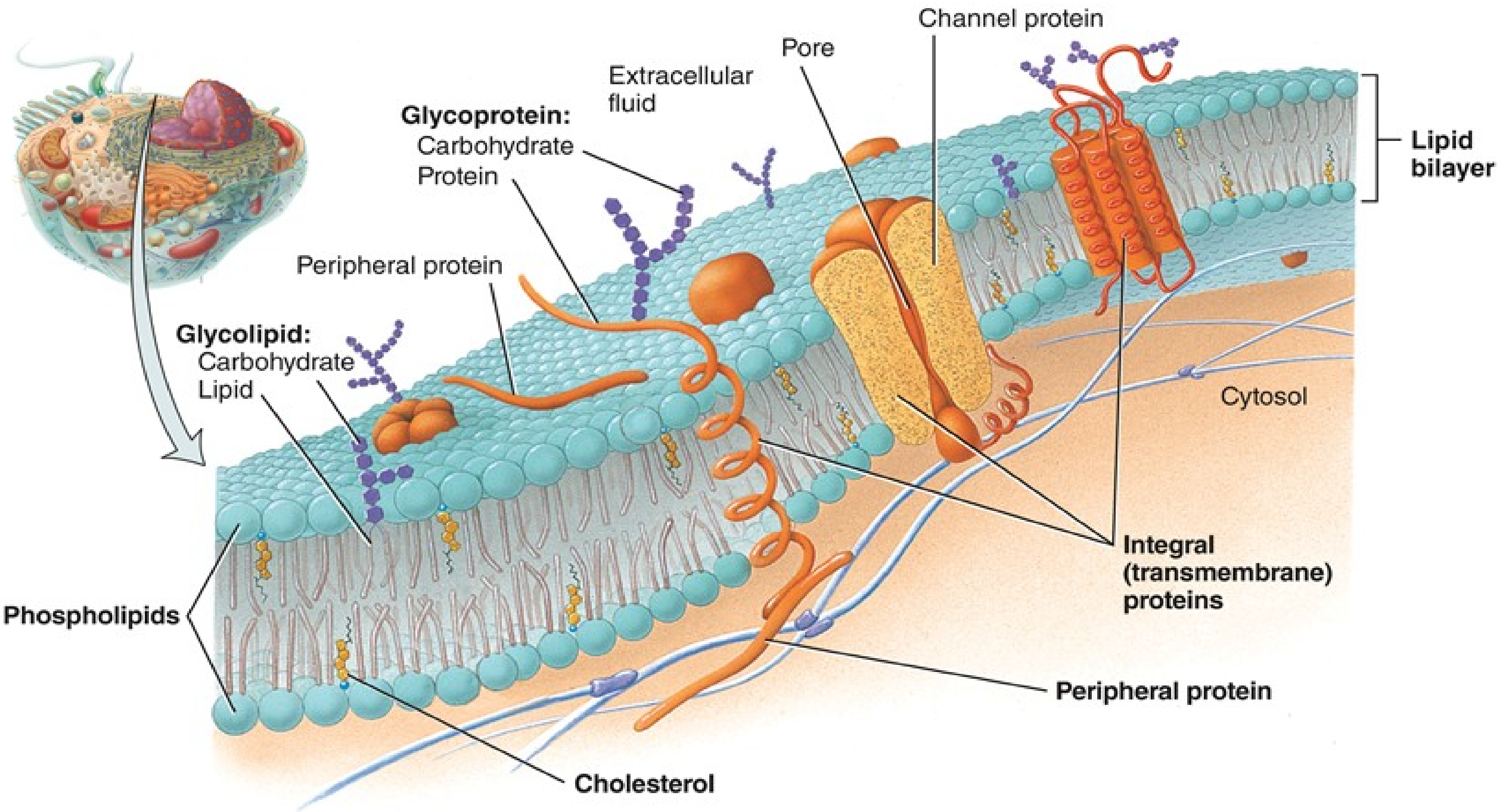
The Plasma Membrane



[Membrane Anatomy Link]

Membrane Proteins

- Two types of membrane proteins are
  - Integral (also called transmembrane) proteins
    - Span the entire width of the membrane
      - Projects into cell's interior; projects outward into extracellular space
  - Peripheral proteins
    - Location on only one side of the membrane

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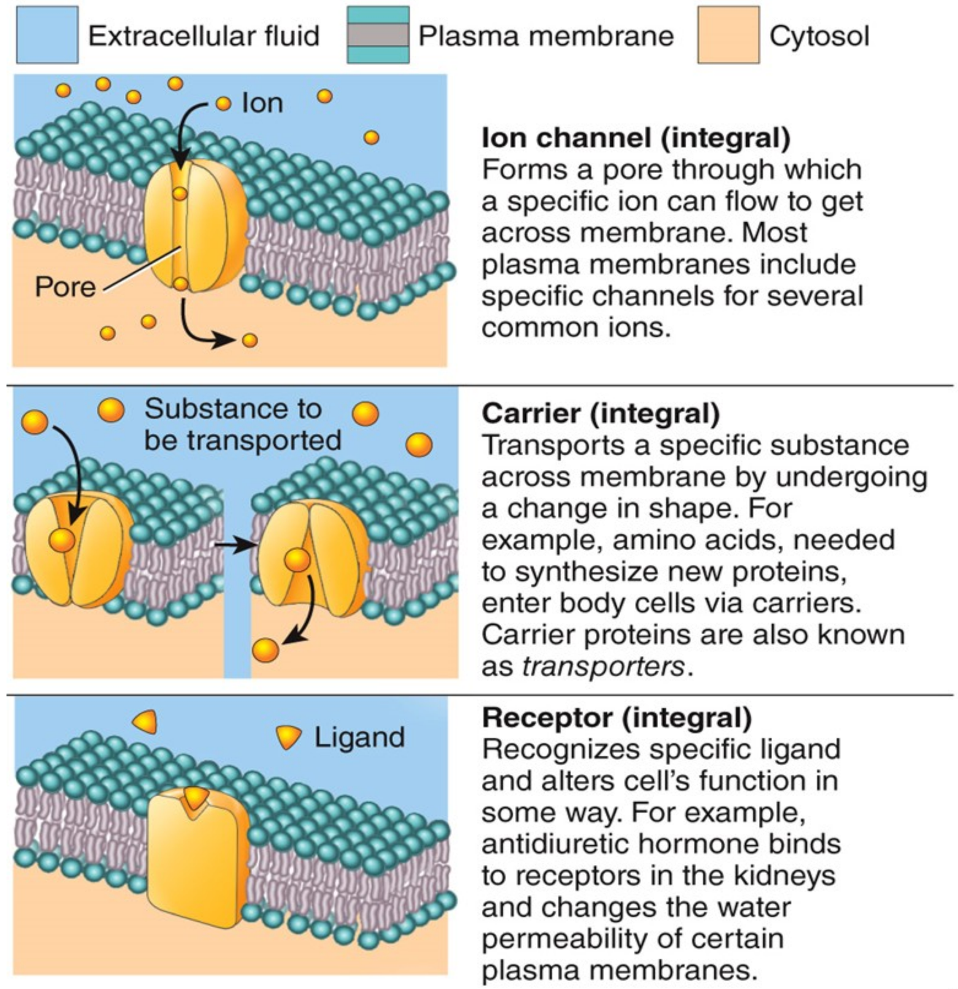
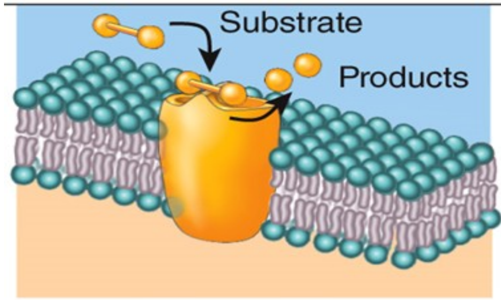
Functions of Membrane Proteins

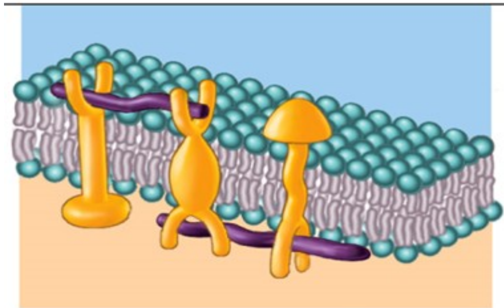


Functions of Membrane Proteins

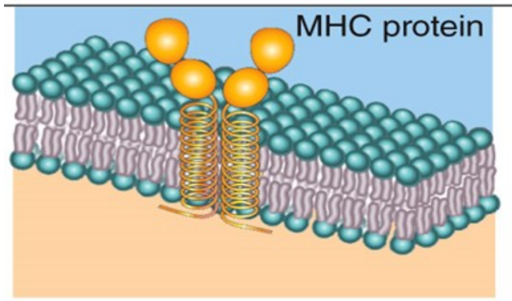
Extracellular fluid | Plasma membrane | Cytosol



Enzyme (integral and peripheral)
Catalyzes reaction inside or outside cell (depending on which direction the active site faces). For example, lactase protruding from epithelial cells lining your small intestine splits the disaccharide lactose in the milk you drink.



Linker (integral and peripheral)
Anchors filaments inside and outside the plasma membrane, providing structural stability and shape for the cell. May also participate in movement of the cell or link two cells together.



Cell identity marker (glycoprotein)
Distinguishes your cells from anyone else's (unless you are an identical twin). An important class of such markers are the major histocompatibility (MHC) proteins.

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Membrane Fluidity

- Membranes are fluid structures because most of the membrane lipids and many of the membrane proteins move easily in the bilayer
  - Membrane lipids and proteins are mobile in their own half of the bilayer
- Cholesterol serves to stabilize the membrane and reduce membrane fluidity

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Membrane Permeability

- Plasma membranes are selectively permeable
  - The lipid bilayer is always permeable to small, nonpolar, uncharged molecules
  - Transmembrane proteins that act as channels or transporters increase the permeability of the membrane
  - Macromolecules are only able to pass through the plasma membrane by vesicular transport

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Gradients Across the Plasma Membrane

- A *concentration gradient* is the difference in the concentration of a chemical between one side of the plasma membrane and the other
- An *electrical gradient* is the difference in concentration of ions between one side of the plasma membrane and the other
- Together, these gradients make up an *electrochemical gradient*

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Transport Across the Plasma Membrane

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Transport Across the Plasma Membrane

- Passive processes – movement from high concentration to low concentration; does not require energy
  - Simple diffusion
  - Facilitated diffusion
  - Osmosis

- Active processes – requires energy
  - Primary and secondary transport
  - Vesicular transport

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Transport Processes Interactions Animation:

Transport Across the Plasma Membrane

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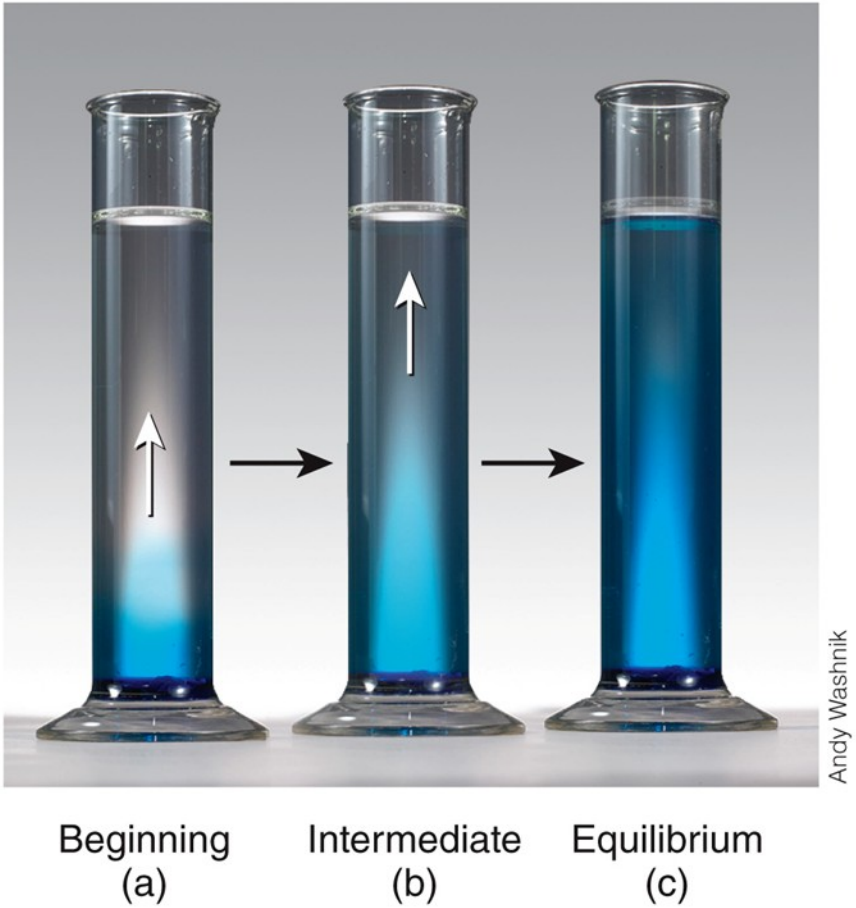
Passive Transport: Simple Diffusion

- Diffusion is influenced by:
  - Steepness of the concentration gradient
  - Temperature
  - Mass of diffusing substance
  - Surface area
  - Diffusion distance



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Passive Transport: Facilitated Diffusion

- Transmembrane proteins help solutes that are too polar or too highly charged move through the lipid bilayer
- The processes involved are:
  - Channel mediated facilitated diffusion
    - Most membrane channels are ion channels
    - Provides a “tunnel”
  - Carrier mediated facilitated diffusion
    - Carriers are transporters
    - Molecule physically binds to transport protein

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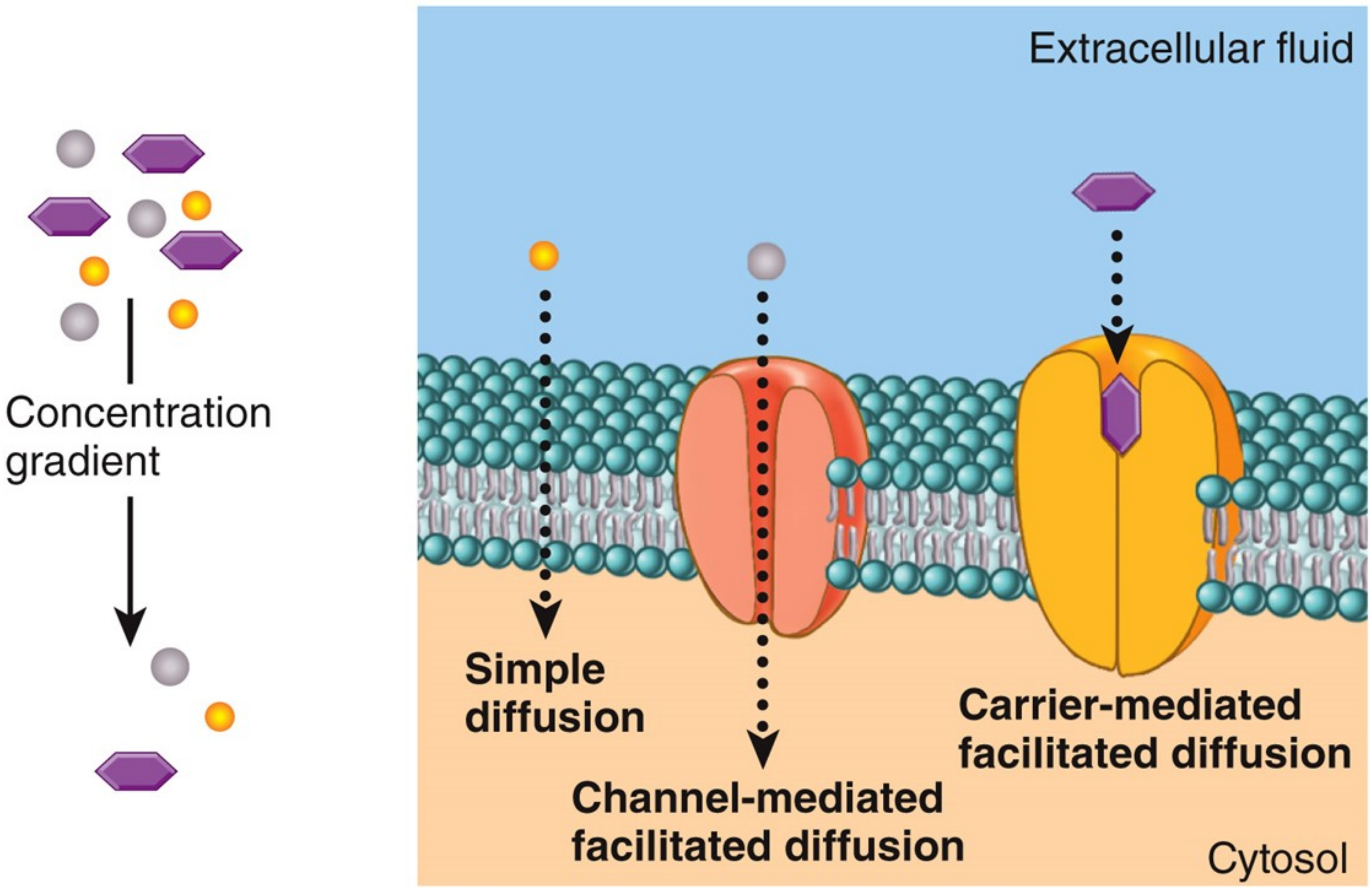
Diffusion: A Comparison



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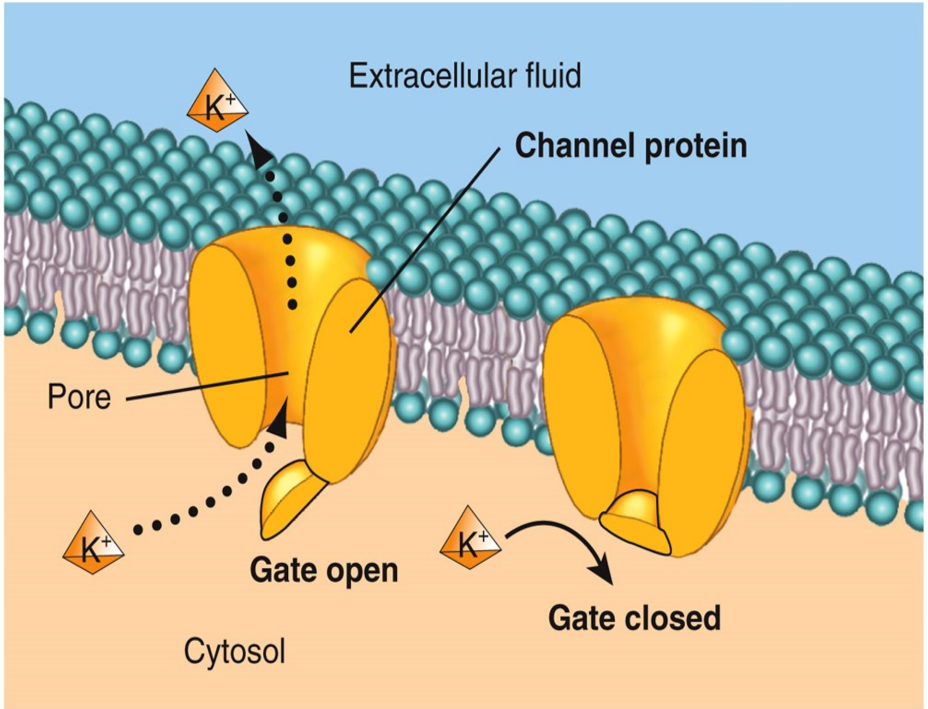
Passive Transport: Channel Mediated Facilitated Diffusion

- Gated channels
  - Part of the protein acts as a “gate” or “plug”, either prohibiting or allowing passage, depending on position



Details of the K^+ channel

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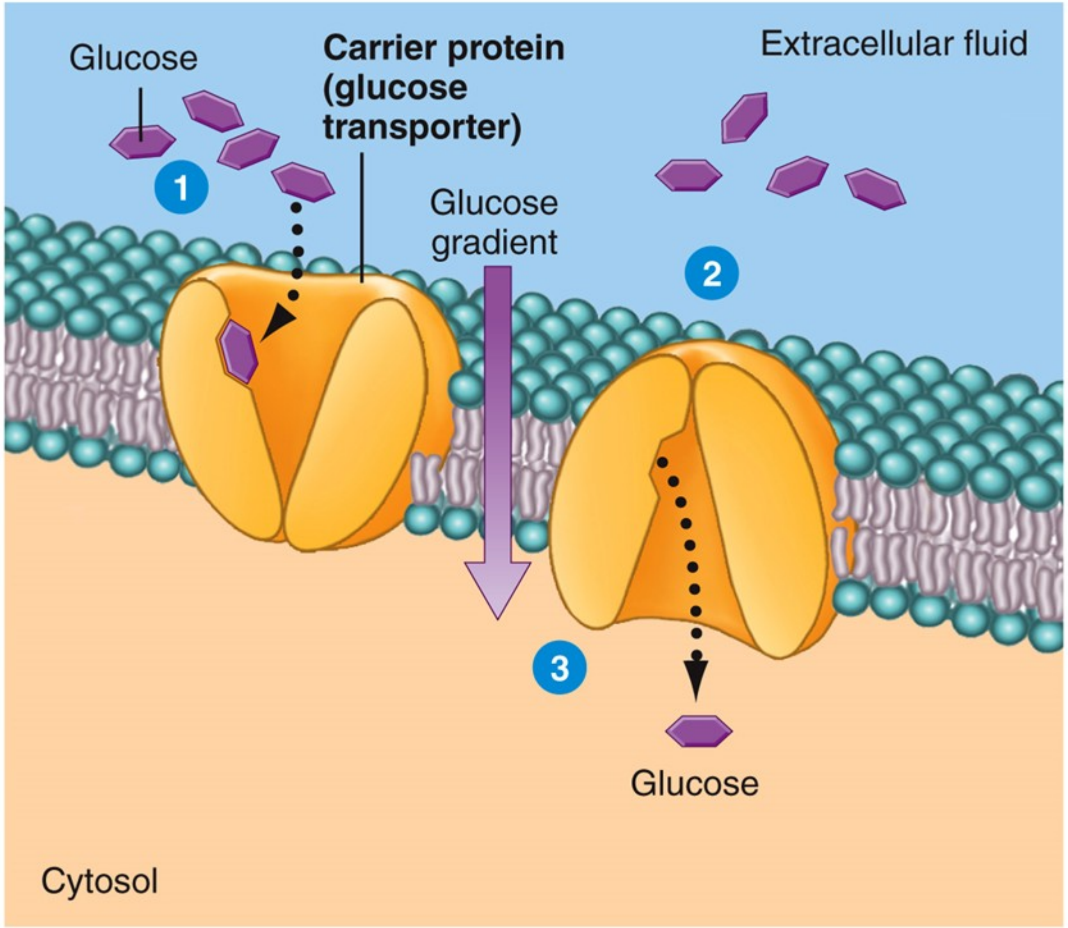
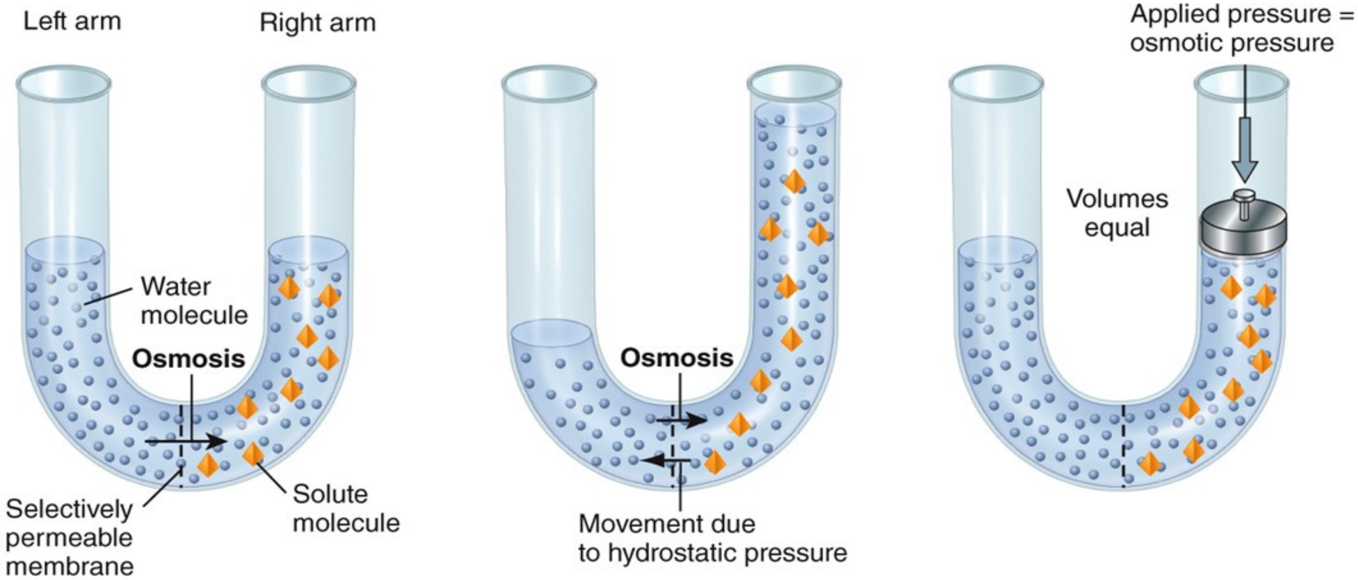
Passive Transport: Channel Mediated Facilitated Diffusion



Passive Transport: Osmosis

- The net movement of water through a selectively permeable membrane from an area of high concentration to an area of low concentration



(a) At start of experiment (b) Equilibrium (c) Restoring starting conditions

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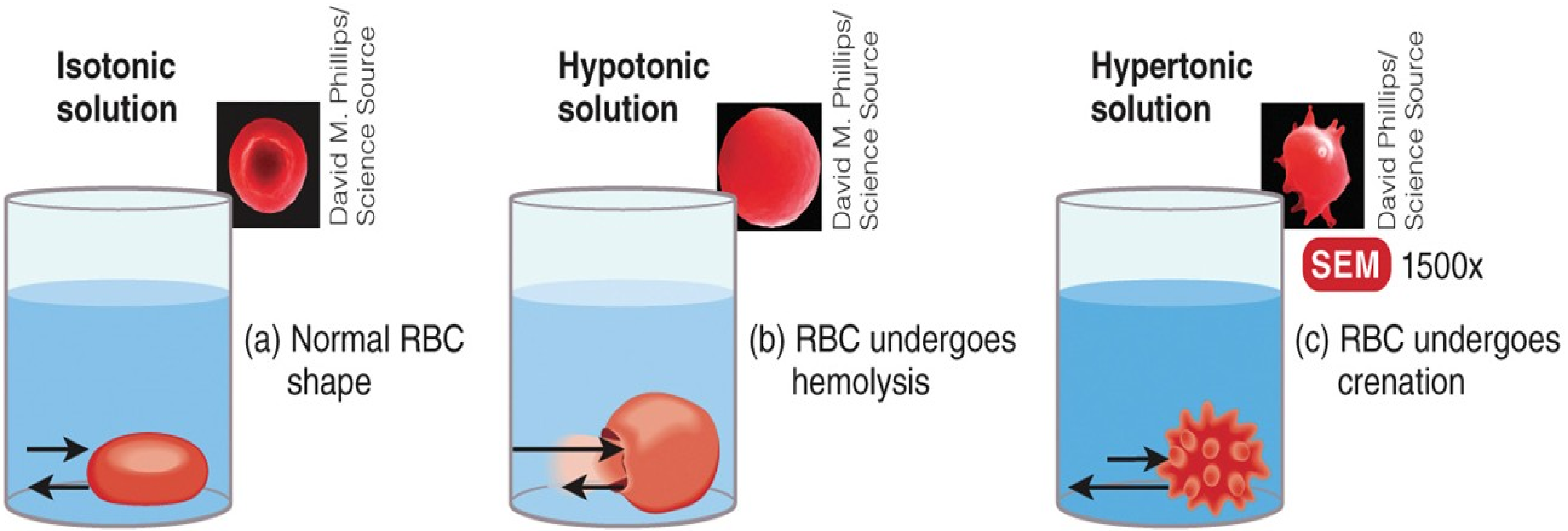
Tonicity

- Tonicity of a solution relates to how the solution influences the shape of body cells



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Tonicity and RBC

Question	Isotonic	Hypotonic	Hypertonic
Is the membrane permeable to water?	Yes	Yes	Yes
Where is the solute concentration higher?	Equal on both sides of cell	Inside the cell	Outside the cell
Where is the solute concentration lower?	Equal on both sides of cell	Outside the cell	Inside the cell
Where is the water concentration higher?	Equal on both sides of cell	Outside the cell	Inside the cell
Where is the water concentration lower?	Equal on both sides of cell	Inside the cell	Outside the cell
Which way will there be net movement of water?	None	Outside to inside	Inside to outside
What will happen to the cell size?	Stay the same	Swell (cell may burst open)	Shrink

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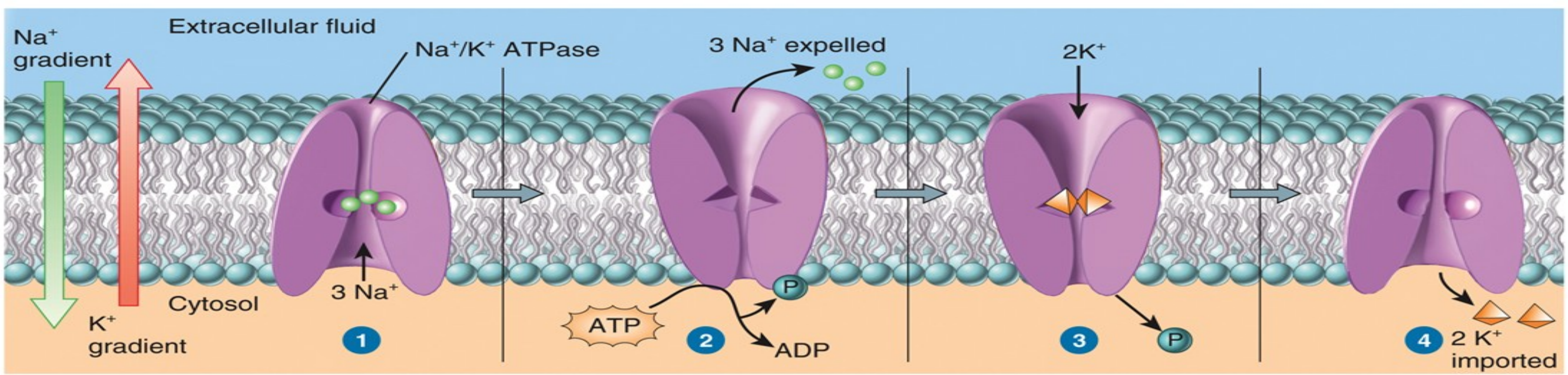
Active Transport: Primary

- Energy derived from ATP changes the shape of a transporter protein which pumps a substance across a plasma membrane against its concentration gradient



1. 3 sodium ions (Na^+) from the cytosol bind to the inside surface of the sodium–potassium pump.
2. Na^+ binding triggers ATP to bind to the pump and be split into ADP and P (phosphate). The energy from ATP splitting causes the protein to change shape, which moves the Na^+ to the outside.
3. 2 potassium ions (K^+) land on the outside surface of the pump and cause the P to be released.
4. The release of the P causes the pump to return to its original shape, which moves the K^+ into the cell.

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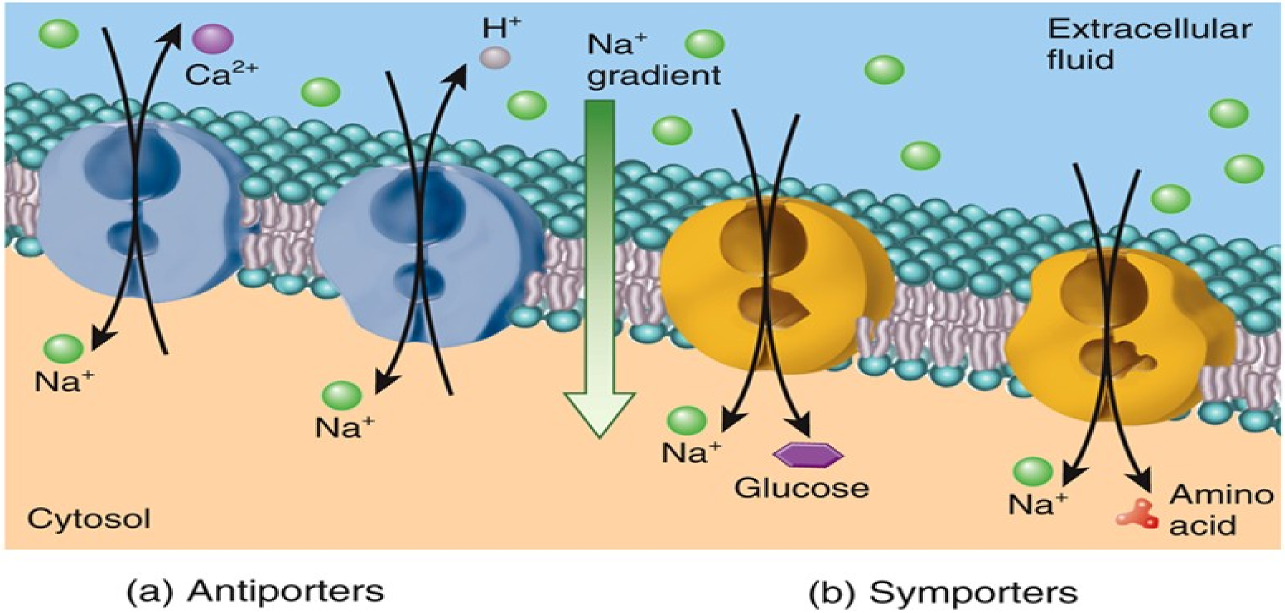
Passive Transport: Secondary

- Stored energy (in a hydrogen or sodium concentration gradient) is used to drive other substances against their own concentration gradients



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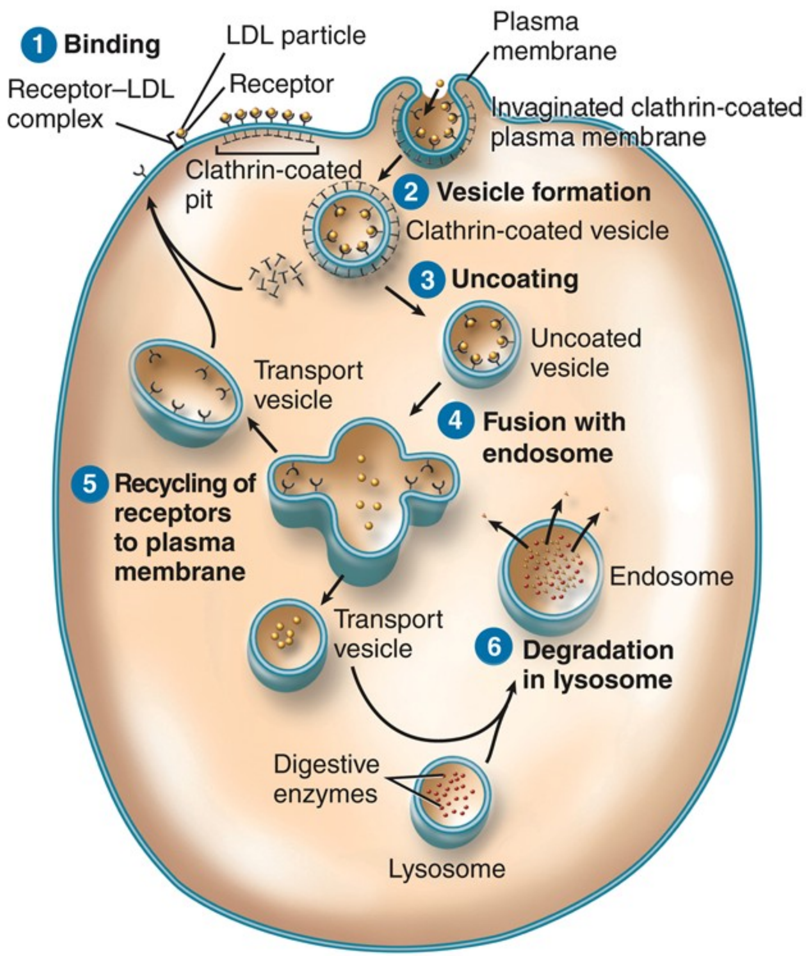
Active Transport in Vesicles: Receptor-Mediated Endocytosis



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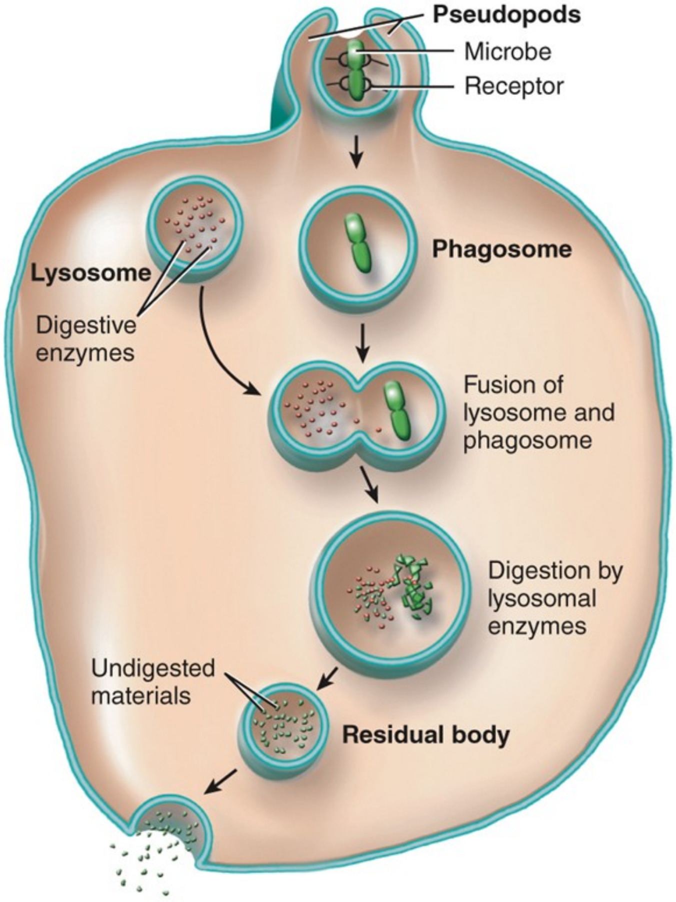
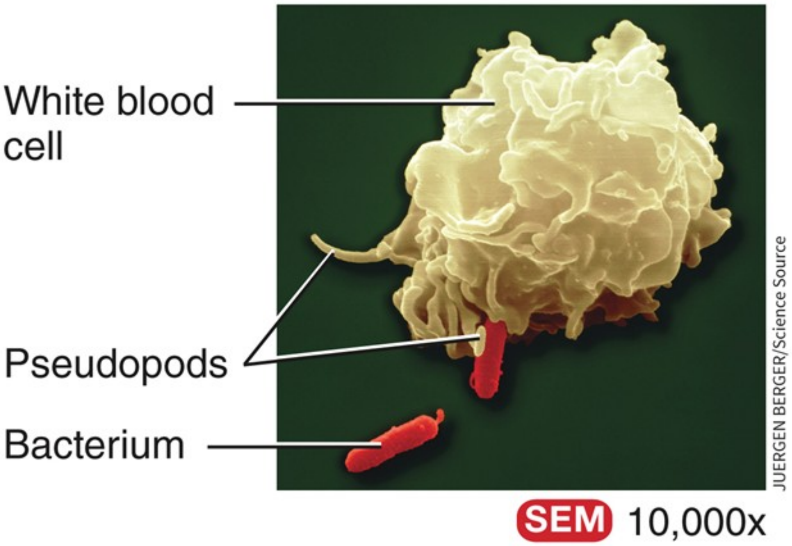
Active Transport in Vesicles: Phagocytosis



(a) Diagram of the process



(b) White blood cell engulfing a bacterium

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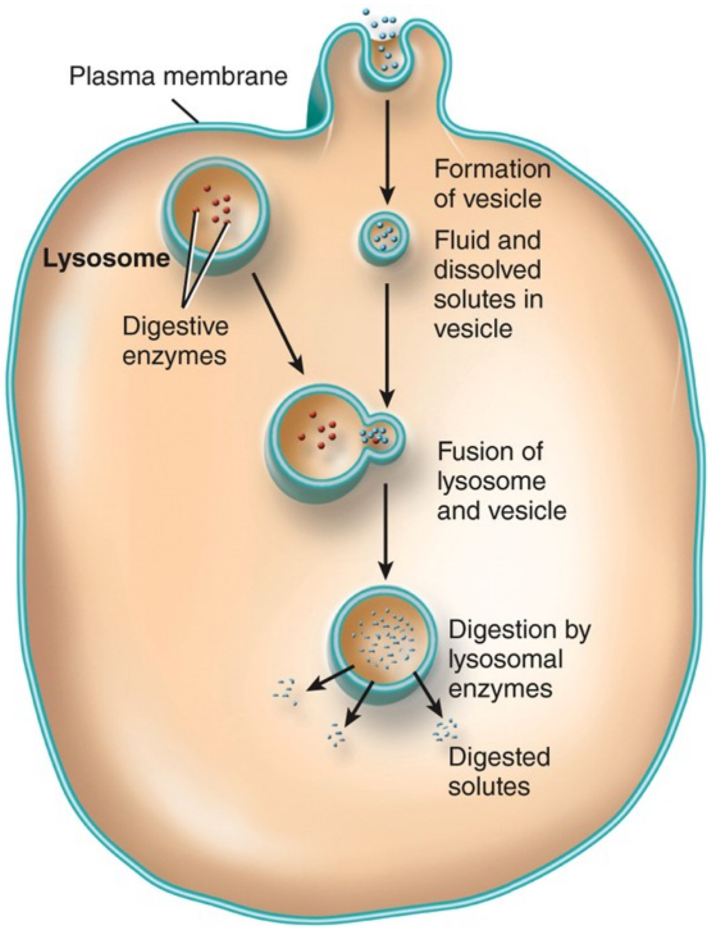
Active Transport in Vesicles: Bulk Phase Endocytosis (Pinocytosis)



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Active Transport in Vesicles: Exocytosis & Transcytosis

- Exocytosis – membrane-enclosed secretory vesicles fuse with the plasma membrane and release their contents into the extracellular fluid

- Transcytosis – a combination of endocytosis and exocytosis used to move substances from one side of a cell, across it, and out the other side

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A Comparison of Transport Types

Transport Process	Description	Substances Transported
PASSIVE PROCESSES	Movement of substances down a concentration gradient until equilibrium is reached; do not require cellular energy in the form of ATP.	
Diffusion	Movement of molecules or ions down a concentration gradient due to their kinetic energy until they reach equilibrium.	
Simple diffusion	Passive movement of a substance down its concentration gradient through the lipid bilayer of the plasma membrane without the help of membrane transport proteins.	Nonpolar, hydrophobic solutes: oxygen, carbon dioxide, and nitrogen gases; fatty acids; steroids; and fat-soluble vitamins. Polar molecules such as water, urea, and small alcohols.

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A Comparison of Transport Types

Transport Process	Description	Substances Transported
Facilitated diffusion	Passive movement of a substance down its concentration gradient through the lipid bilayer by transmembrane proteins that function as channels or carriers.	Polar or charged solutes: glucose; fructose; galactose; some vitamins; and ions such as K^+, Cl^-, Na^+, and Ca^{2+}.
Osmosis	Passive movement of water molecules across a selectively permeable membrane from an area of higher to lower water concentration until equilibrium is reached.	Solvent: water in living systems.

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A Comparison of Transport Types

Transport Process	Description	Substances Transported
ACTIVE PROCESSES	Movement of substances against a concentration gradient; requires cellular energy in the form of ATP.	
Active Transport	Active process in which a cell expends energy to move a substance across the membrane against its concentration gradient by transmembrane proteins that function as carriers.	Polar or charged solutes.
Primary active transport	Active process in which a substance moves across the membrane against its concentration gradient by pumps (carriers) that use energy supplied by hydrolysis of ATP.	Na^+, K^+, Ca^{2+}, H^+, I^-, Cl^-, and other ions.
Secondary active transport	Coupled active transport of two substances across the membrane using energy supplied by a Na^+ or H^+ concentration gradient maintained by primary active transport pumps. Antiporters move Na+ (or H^+) and another substance in opposite directions across the membrane; symporters move Na^+ (or H^+) and another substance in the same direction across the membrane.	Antiport: Ca^{2+}, H^+ out of cells. Symport: glucose, amino acids into cells.

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A Comparison of Transport Types

Transport Process	Description	Substances Transported
Transport in Vesicles	Active process in which substances move into or out of cells in vesicles that bud from plasma membrane; requires energy supplied by ATP.	
Endocytosis	Movement of substances into a cell in vesicles.	
Receptor-mediated endocytosis	Ligand–receptor complexes trigger infolding of a clathrin-coated pit that forms a vesicle containing ligands.	Ligands: transferrin, low-density lipoproteins (LDLs), some vitamins, certain hormones, and antibodies.
Phagocytosis	“Cell eating”; movement of a solid particle into a cell after pseudopods engulf it to form a phagosome.	Bacteria, viruses, and aged or dead cells.
Bulk-phase endocytosis	“Cell drinking”; movement of extracellular fluid into a cell by infolding of plasma membrane to form a vesicle.	Solutes in extracellular fluid.

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A Comparison of Transport Types

Transport Process	Description	Substances Transported
Exocytosis	Movement of substances out of a cell in secretory vesicles that fuse with the plasma membrane and release their contents into the extracellular fluid.	Neurotransmitters, hormones, and digestive enzymes.
Transcytosis	Movement of a substance through a cell as a result of endocytosis on one side and exocytosis on the opposite side.	Substances, such as antibodies, across endothelial cells. This is a common route for substances to pass between blood plasma and interstitial fluid.

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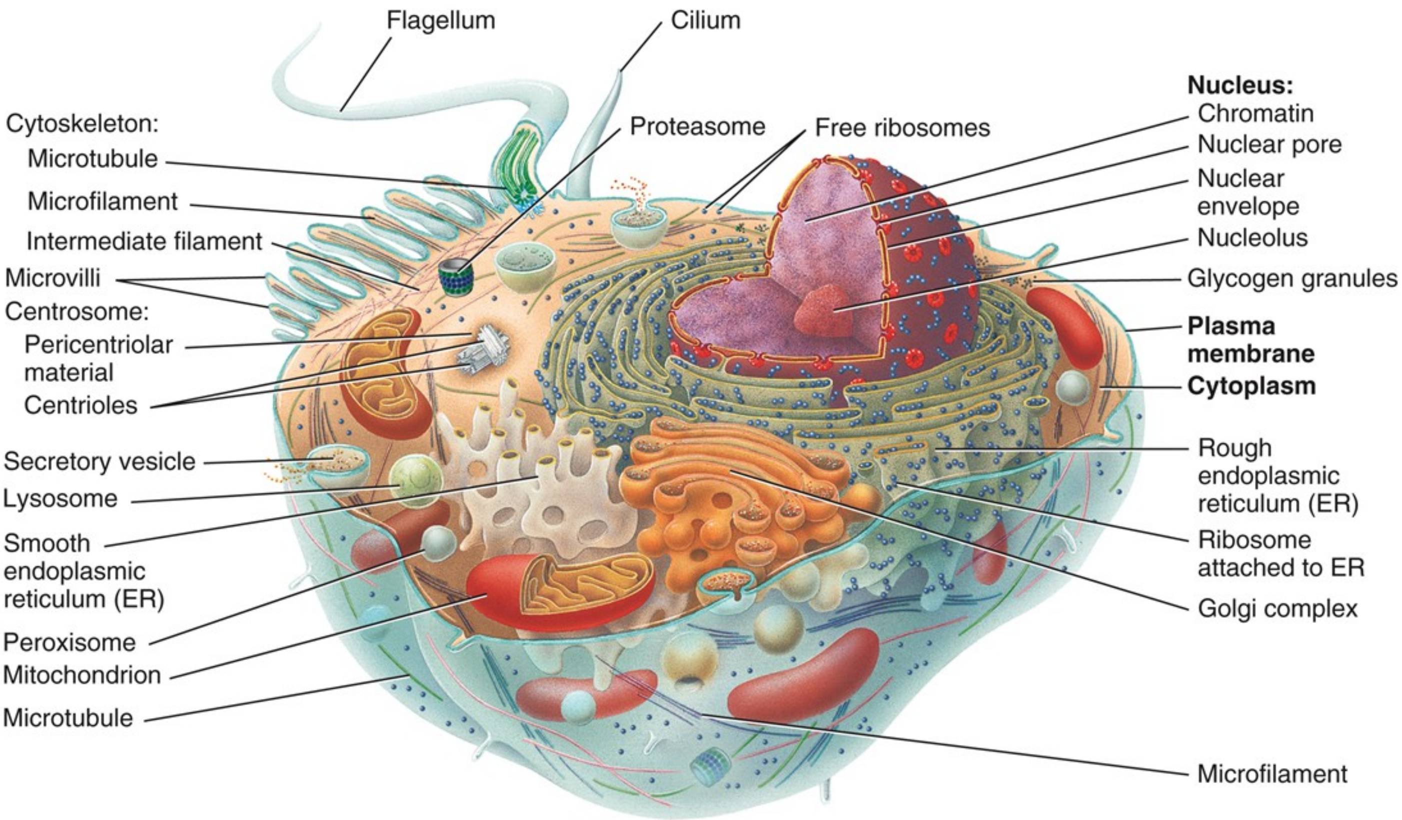
Structures of a Cell



Sectional view

[Cell Anatomy Link]

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Cytoplasm

- The cytoplasm has 2 components:
  - Cytosol - also known as the intracellular fluid portion of the cytoplasm
  - Organelles - the specialized structures that have specific shapes and perform specific functions

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Cell Parts and Their Functions

Part	Description	Functions
PLASMA MEMBRANE	Fluid mosaic lipid bilayer (phospholipids, cholesterol, and glycolipids) studded with proteins; surrounds cytoplasm.	Protects cellular contents; makes contact with other cells; contains channels, transporters, receptors, enzymes, cell-identity markers, and linker proteins; mediates entry and exit of substances.
CYTOPLASM	Cellular contents between plasma membrane and nucleus—cytosol and organelles.	Site of all intracellular activities except those occurring in the nucleus.
Cytosol	Composed of water, solutes, suspended particles, lipid droplets, and glycogen granules.	Fluid in which many of cell's metabolic reactions occur.
	The cytoskeleton is a network in the cytoplasm composed of three protein filaments: microfilaments, intermediate filaments, and microtubules.	The cytoskeleton maintains shape and general organization of cellular contents; responsible for cell movements.

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Cell Parts and Their Functions

Part	Description	Functions
Organelles	Specialized structures with characteristic shapes.	Each organelle has specific functions.
Centrosome	Pair of centrioles plus pericentriolar matrix.	The pericentriolar matrix contains tubulins, which are used for growth of the mitotic spindle and microtubule formation.
Cilia and flagella	Motile cell surface projections that contain 20 microtubules and a basal body.	Cilia: move fluids over cell's surface; flagella: move entire cell.
Ribosome	Composed of two subunits containing ribosomal RNA and proteins; may be free in cytosol or attached to rough ER.	Protein synthesis.

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Cell Parts and Their Functions

Part	Description	Functions
Endoplasmic reticulum (ER)	Membranous network of flattened sacs or tubules. Rough ER is covered by ribosomes and is attached to the nuclear envelope; smooth ER lacks ribosomes.	Rough ER: synthesizes glycoproteins and phospholipids that are transferred to cellular organelles, inserted into plasma membrane, or secreted during exocytosis; smooth ER: synthesizes fatty acids and steroids, inactivates or detoxifies drugs, removes phosphate group from glucose-6-phosphate, and stores and releases calcium ions in muscle cells.
Golgi complex	Consists of 3–20 flattened membranous sacs called saccules; structurally and functionally divided into entry (*cis*) face, medial saccules, and exit (*trans*) face.	Entry (*cis*) face accepts proteins from rough ER; medial saccules form glycoproteins, glycolipids, and lipoproteins; exit (*trans*) face modifies molecules further, then sorts and packages them for transport to their destinations.

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Cell Parts and Their Functions

Part	Description	Functions
Lysosome	Vesicle formed from Golgi complex; contains digestive enzymes.	Fuses with and digests contents of endosomes, phagosomes, and vesicles formed during bulk-phase endocytosis and transports final products of digestion into cytosol; digests worn-out organelles (autophagy), entire cells (autolysis), and extracellular materials.
Peroxisome	Vesicle containing oxidases (oxidative enzymes) and catalase (decomposes hydrogen peroxide); new peroxisomes bud from preexisting ones.	Oxidizes amino acids and fatty acids; detoxifies harmful substances, such as hydrogen peroxide and associated free radicals.

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Cell Parts and Their Functions

Part	Description	Functions
Proteasome	Tiny barrel-shaped structure that contains proteases (proteolytic enzymes).	Degrades unneeded, damaged, or faulty proteins by cutting them into small peptides.
Mitochondrion	Consists of an external and an internal mitochondrial membrane, mitochondrial cristae, and mitochondrial matrix; new mitochondria form from preexisting ones.	Site of aerobic cellular respiration reactions that produce most of a cell's ATP. Plays an important early role in apoptosis.

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Cell Parts and Their Functions

Part	Description	Functions
NUCLEUS	Consists of a nuclear envelope with pores, nucleoli, and chromosomes, which exist as a tangled mass of chromatin in interphase cells.	Nuclear pores control the movement of substances between the nucleus and cytoplasm, nucleoli produce ribosomes, and chromosomes consist of genes that control cellular structure and direct cellular functions.

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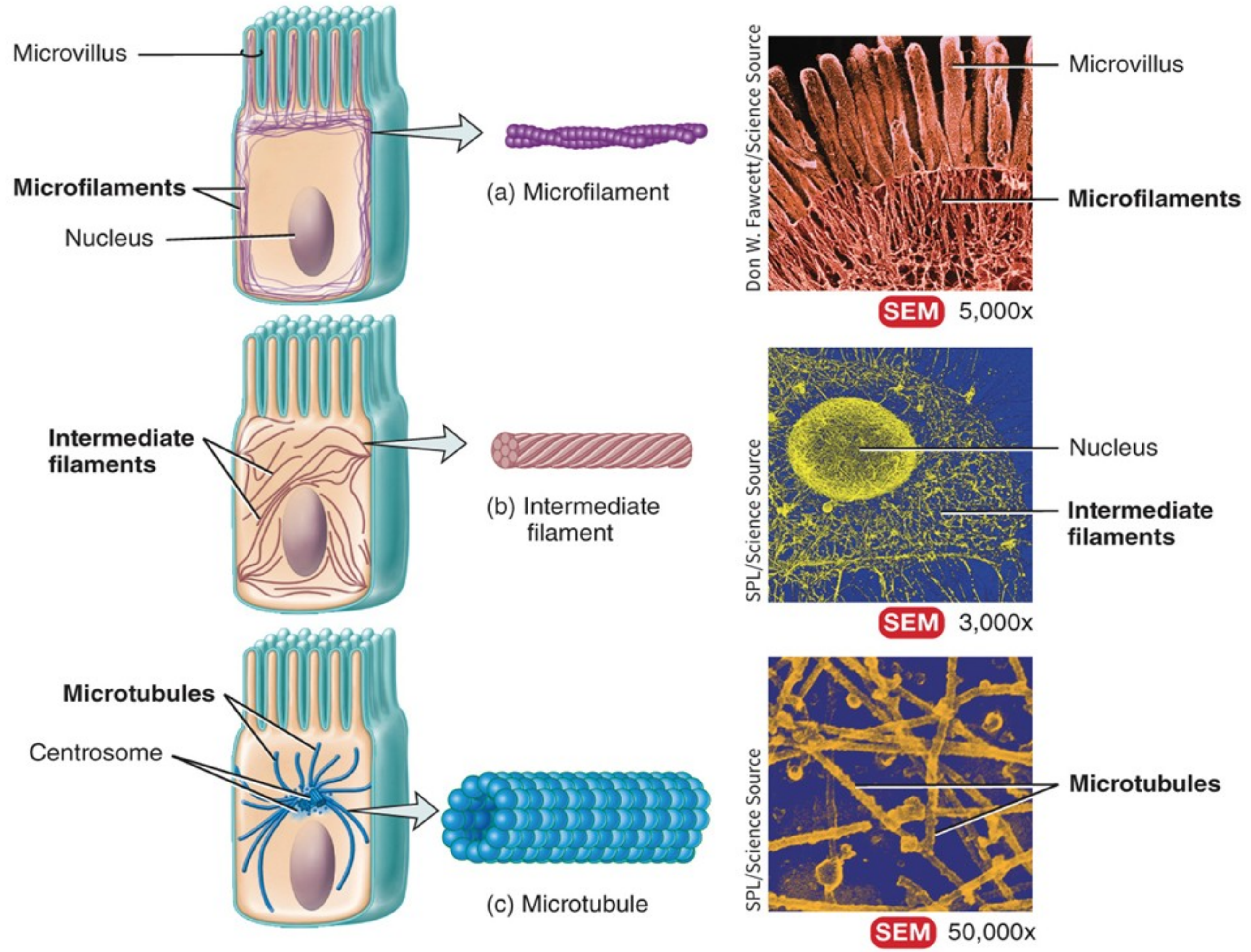
Cytoskeleton



Cytoskeleton

- Microfilaments
  - Composed of actin or myosin proteins subunits
  - Can change length
  - Involved with cell division and cell movement
- Intermediate filaments
  - Composed of collagen or keratin
  - Do not change length
  - Provide structural support

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Cytoskeleton

- Microtubules
  - Composed of tubulin protein subunits
  - Can change length
  - Involved with chromosome movement during cell division

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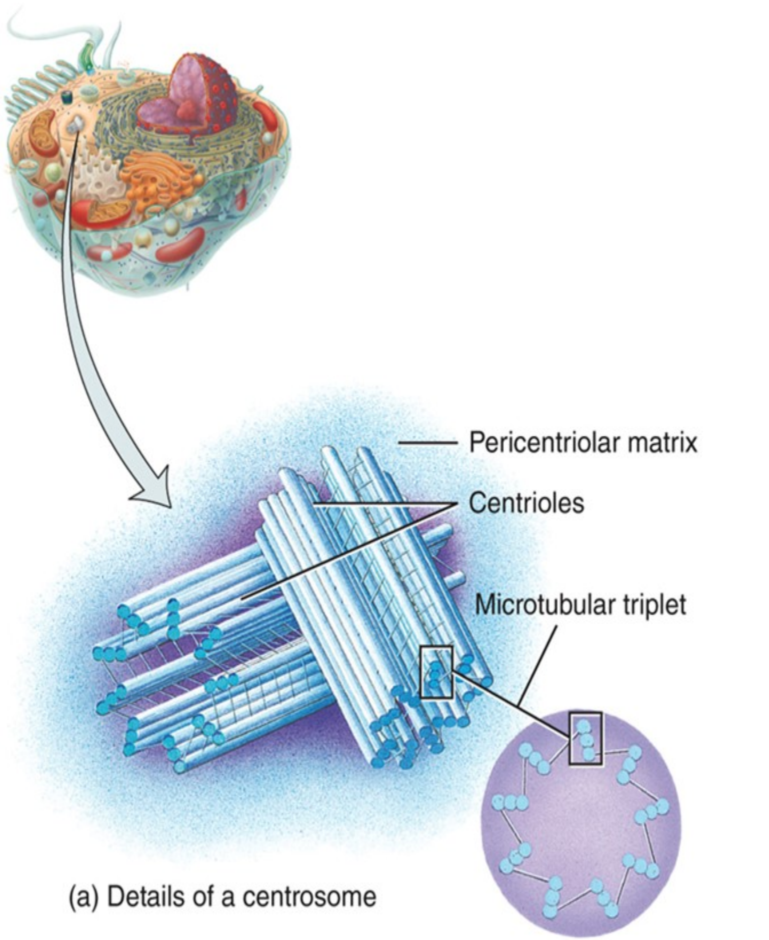
Centrosome/Centrioles

- Centrioles are present in a pair, oriented at right angles to one another
  - Centrosome = pair of centrioles
- Serve as the base of flagella and cilia
  - Called a basal body
- Help with organization of microtubules during cell division



(a) Details of a centrosome

(b) Arrangement of microtubules in centrosome

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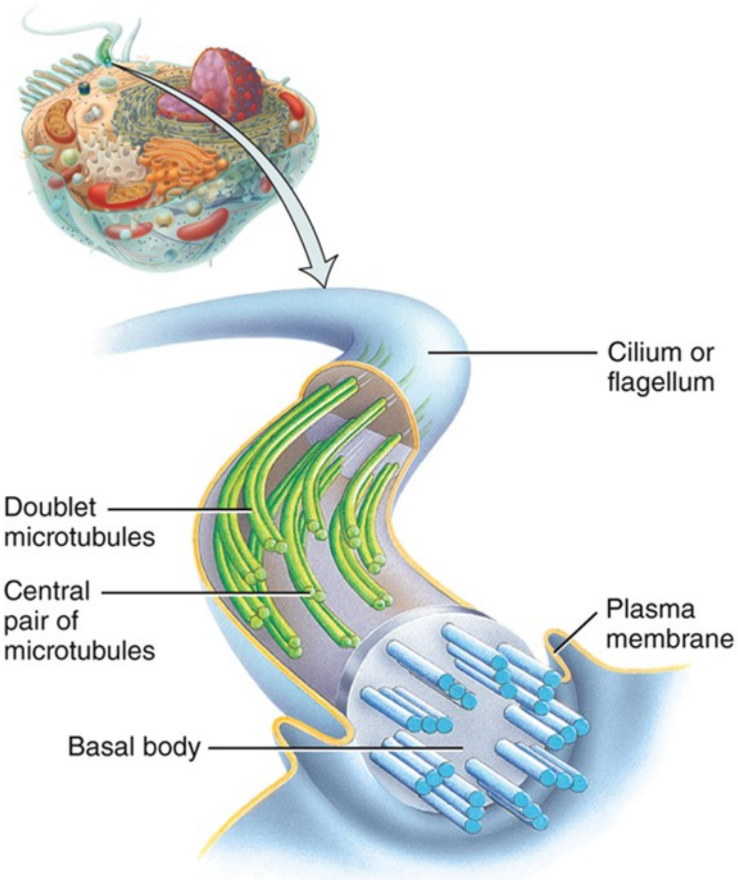
Cilia and Flagella

- Cilia movement helps sweep substances across the surface of the cell
- Flagella aids in location
  - Sperm is the only human cell that has flagella



(a) Arrangement of microtubules in a cilium or flagellum

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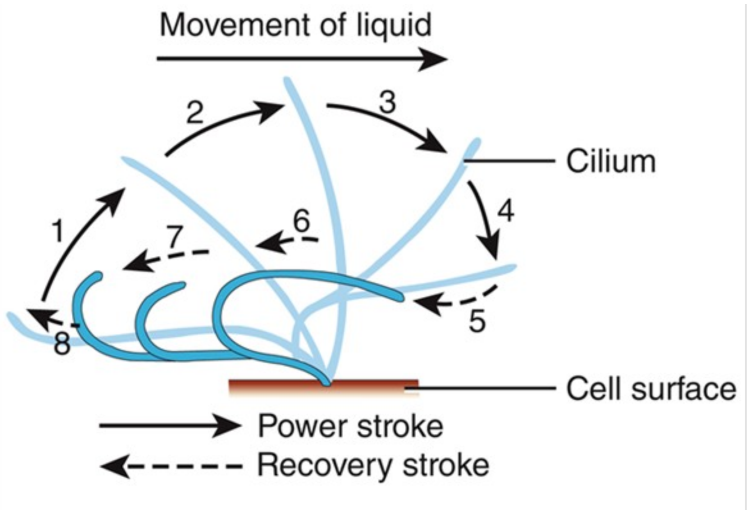
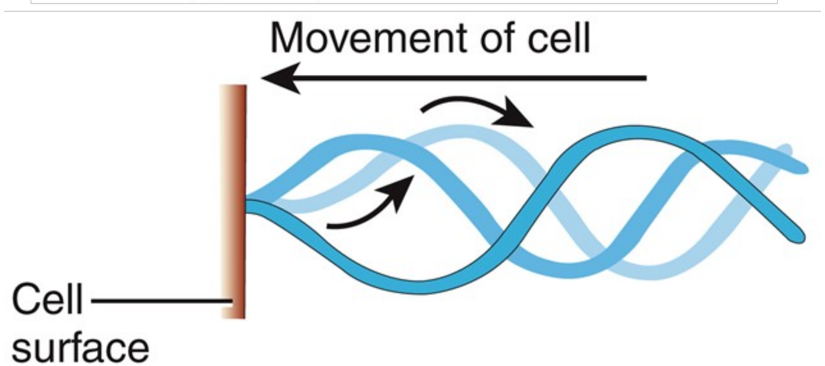
Cilia and Flagella



(d) Ciliary movement



(e) Flagellar movement

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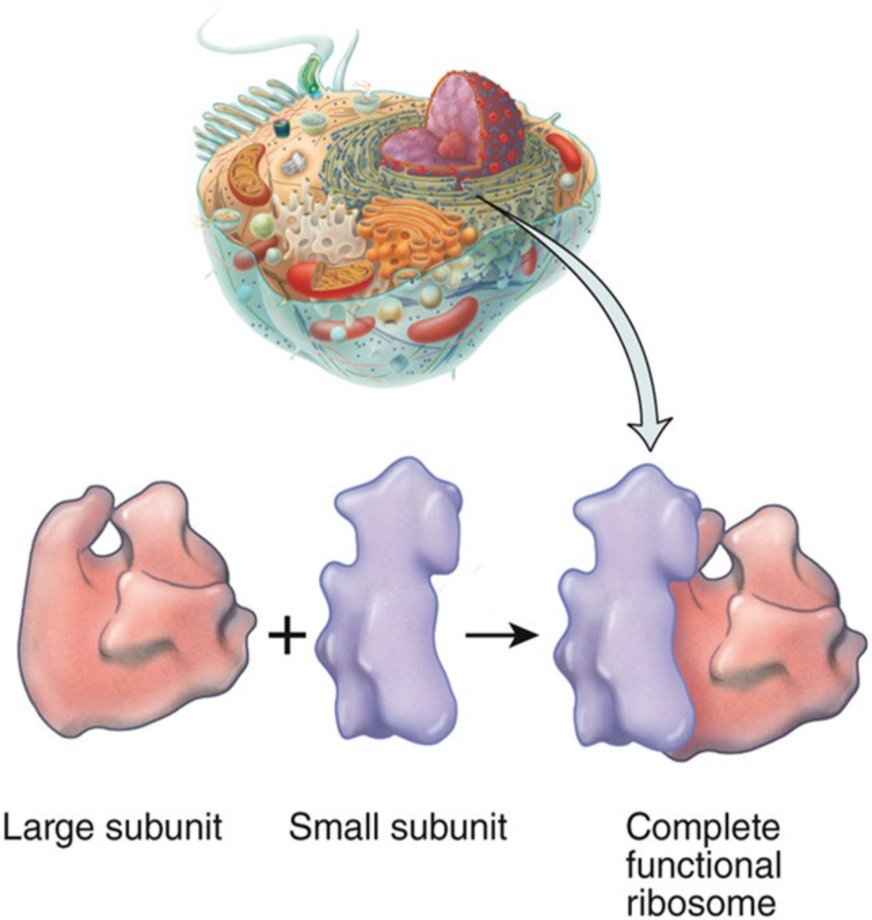
Ribosomes

- Consists of two subunits (small and large)
- Composed of ribosomal RNA (rRNA) and protein
- Site of polypeptide synthesis
  - Location where amino acids are linked together



(a) Details of ribosomal subunits

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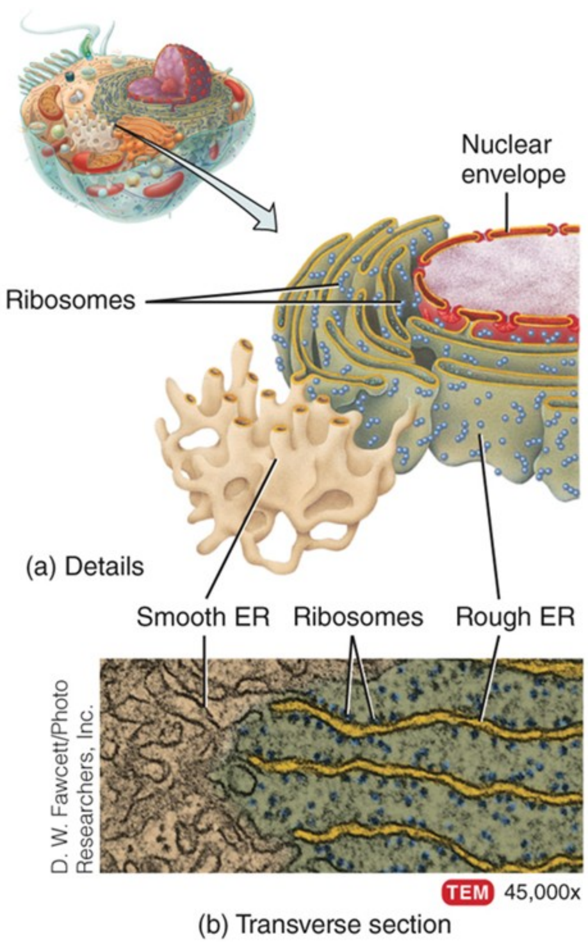
Endoplasmic Reticulum – rough

- Appearance due to being studded with ribosomes
- Part of its membrane can pinch off, forming a transport vesicle that contains the newly-synthesized polypeptide
    - Vesicle fuses with plasma membrane for protein export or fuse with Golgi for proteins to become modified



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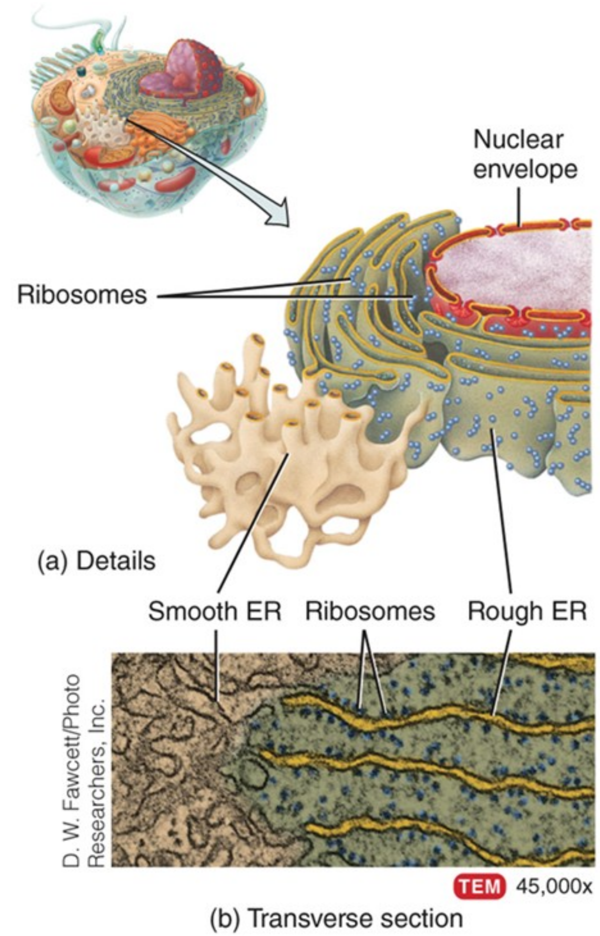
Endoplasmic Reticulum – smooth

- Is not associated with ribosomes
- Involved in storage for ions
    - Example: calcium storage in muscle cells
- Produces lipids
- Associated with enzymes involved with detoxification for certain drugs



(b) Transverse section

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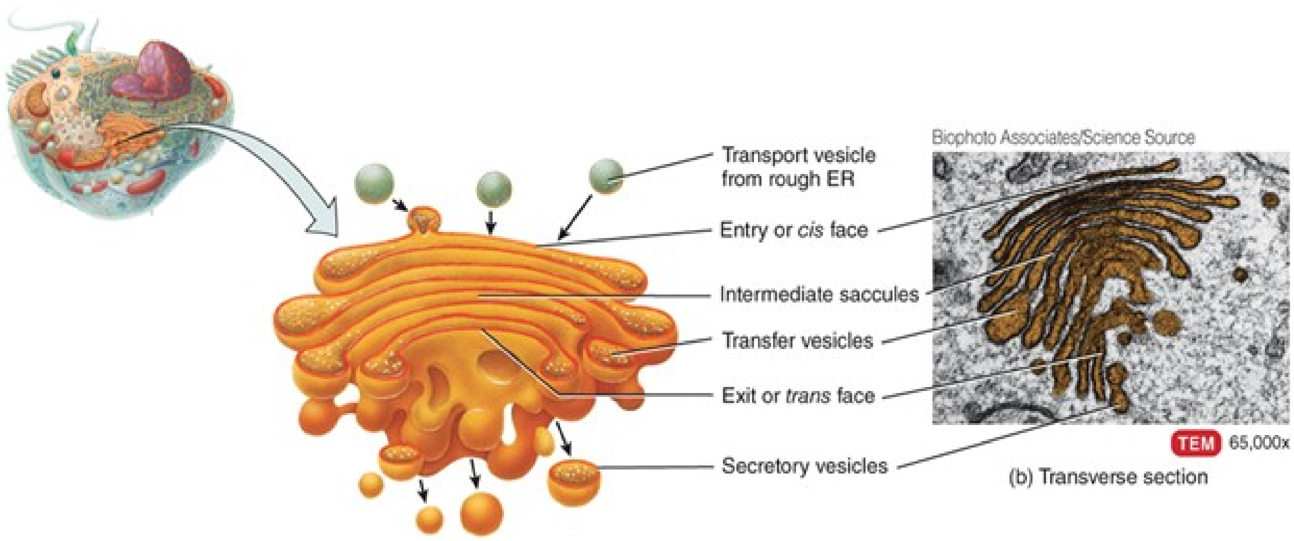
Golgi Complex

- Modifies proteins and packages them for export
  - Parts of the membrane can pinch off, forming a transport vesicle with the proteins in the interior



(a) Details
(b) Transverse section

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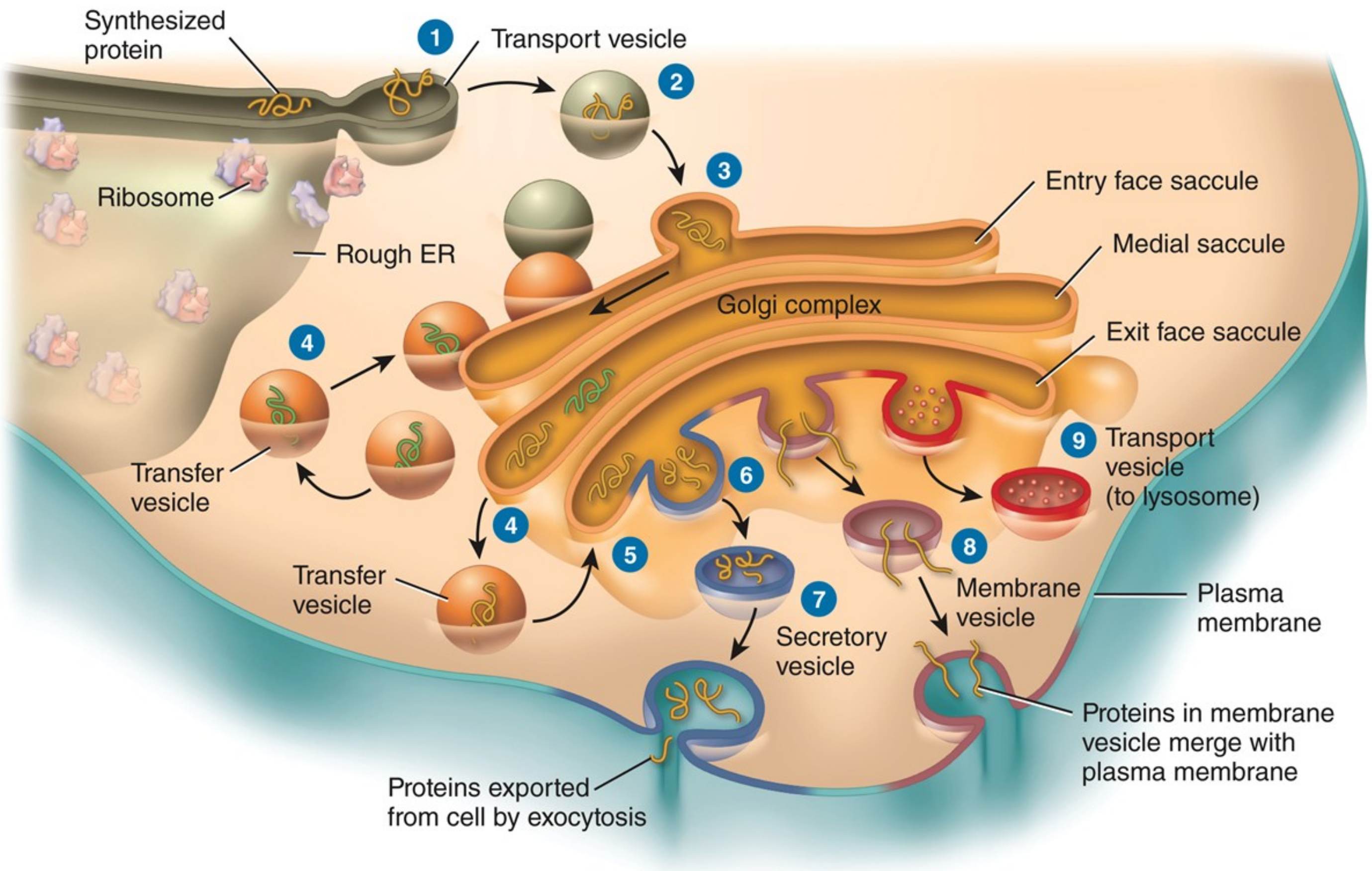
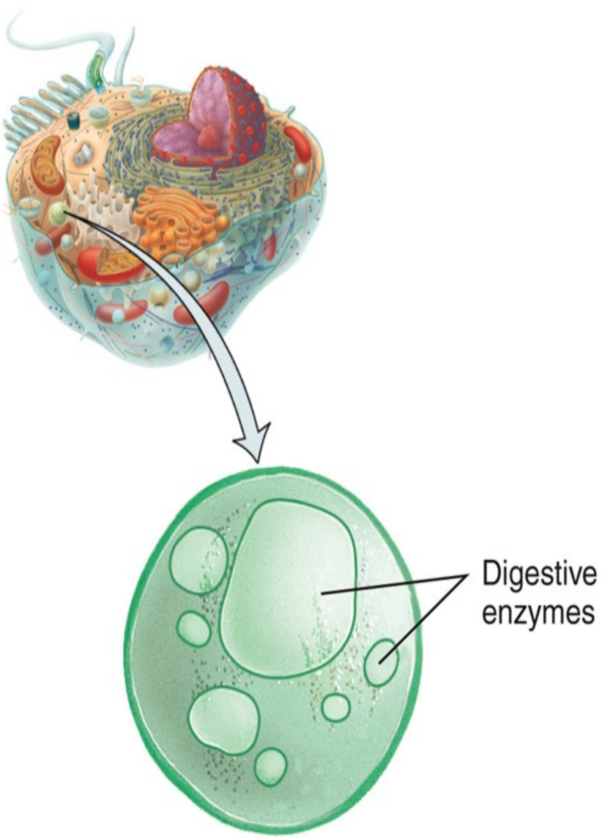
Golgi Complex



Lysosomes

- Membrane-bound packages of hydrolytic/ digestive enzymes
- Fuses with a food vacuole
    - Enzymes can digest the contents and smaller molecules can be transported though membrane into cytosol
- Fuses with membrane-bound package containing old organelles



(a) Lysosome

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Lysosomes

- Function via 60 different digestive and hydrolytic enzymes
  - “Auto-” = self
  - Autophagy
    - Digestion of worn-out organelles by lysosomal enzymes
    - Organelle fuses with lysosome to make autophagosome
  - Autolysis
    - Destruction of entire cell by lysosomal enzymes

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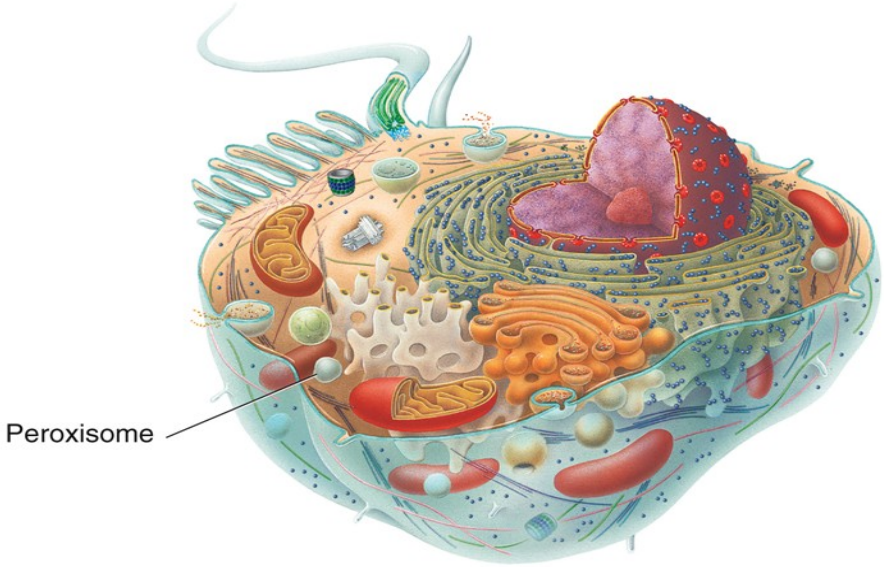
Peroxisomes

- Peroxisomes are structures that are similar in shape to lysosomes, but are smaller and contain enzymes that use oxygen to oxidize (break down) organic substances



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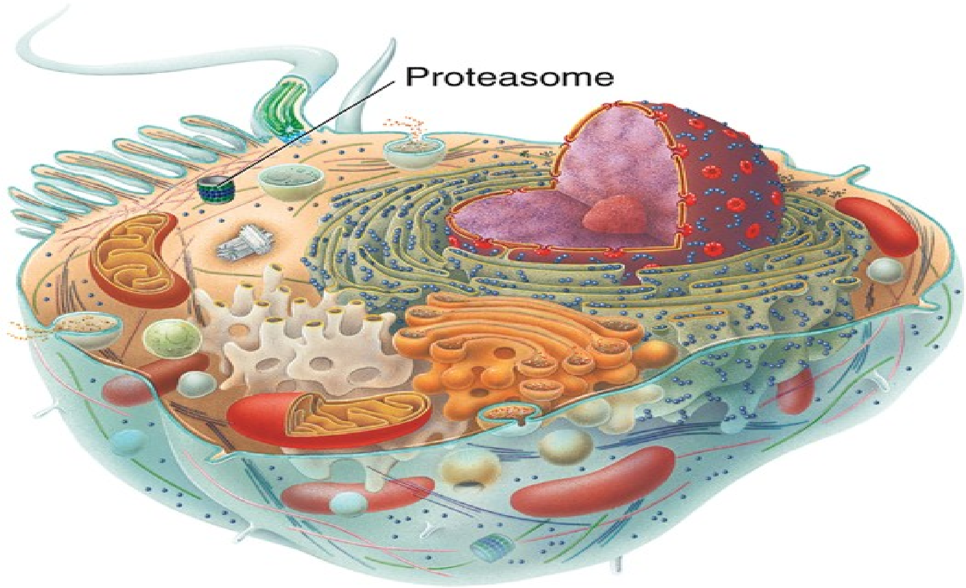
Proteasomes

- Proteasomes are barrel-shaped structures that destroy unneeded, damaged, or faulty proteins by cutting long proteins into smaller peptides



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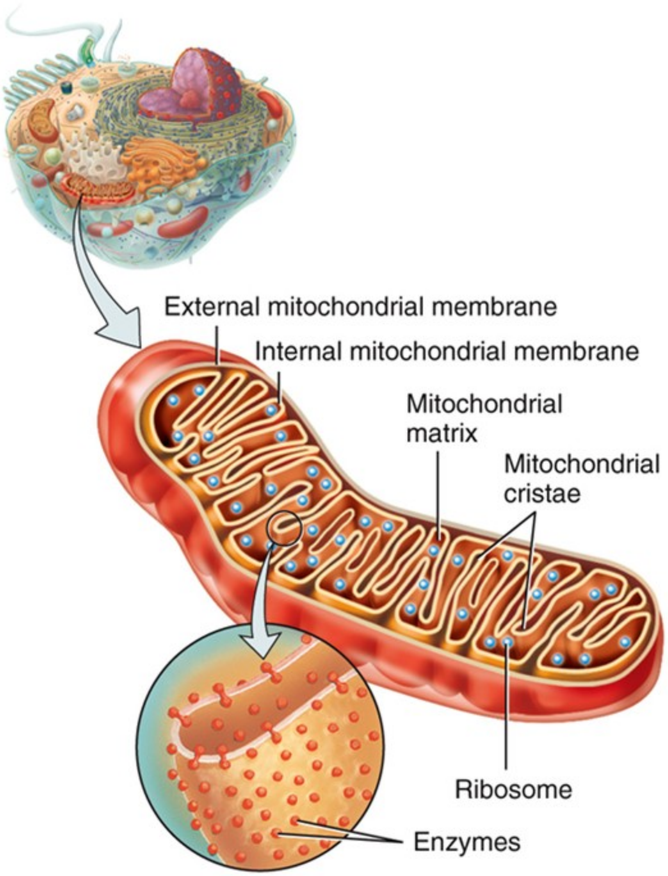
Mitochondria

- Double-membraned
  - Inner membrane has folds called cristae that increases surface area
- Contains its own DNA and ribosomes
  - Mitochondrial DNA is only maternal in origin
- “powerhouse of the cell”
  - Most of the cell’s ATP is produced within the mitochondria during aerobic respiration



(a) Details

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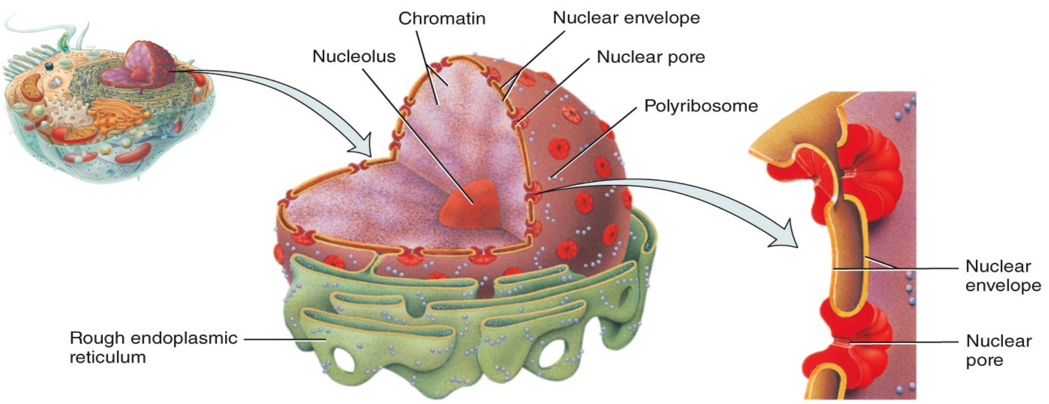
Nucleus

- “control center of cell”
- Contains chromatin
  - DNA and associated histone proteins
  - Called chromosomes when at its most condensed
- Contains nucleolus
  - Composed of DNA, RNA, and protein
  - Site of rRNA synthesis and ribosome assembly



(a) Details of the nucleus (b) Details of the nuclear envelope

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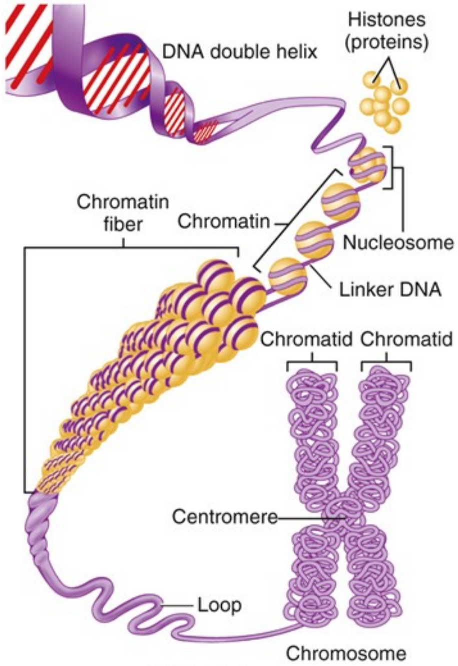
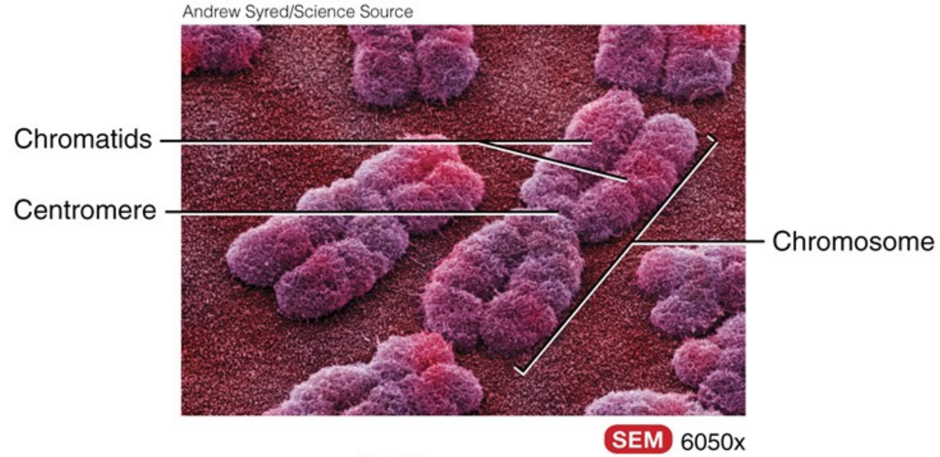
Nucleus

- The nucleus contains the hereditary units of the cell, called genes



(a) Illustration

- Genes are arranged along chromosomes
  - A specific DNA sequence that codes for a polypeptide



(b) Chromosome

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Protein Synthesis

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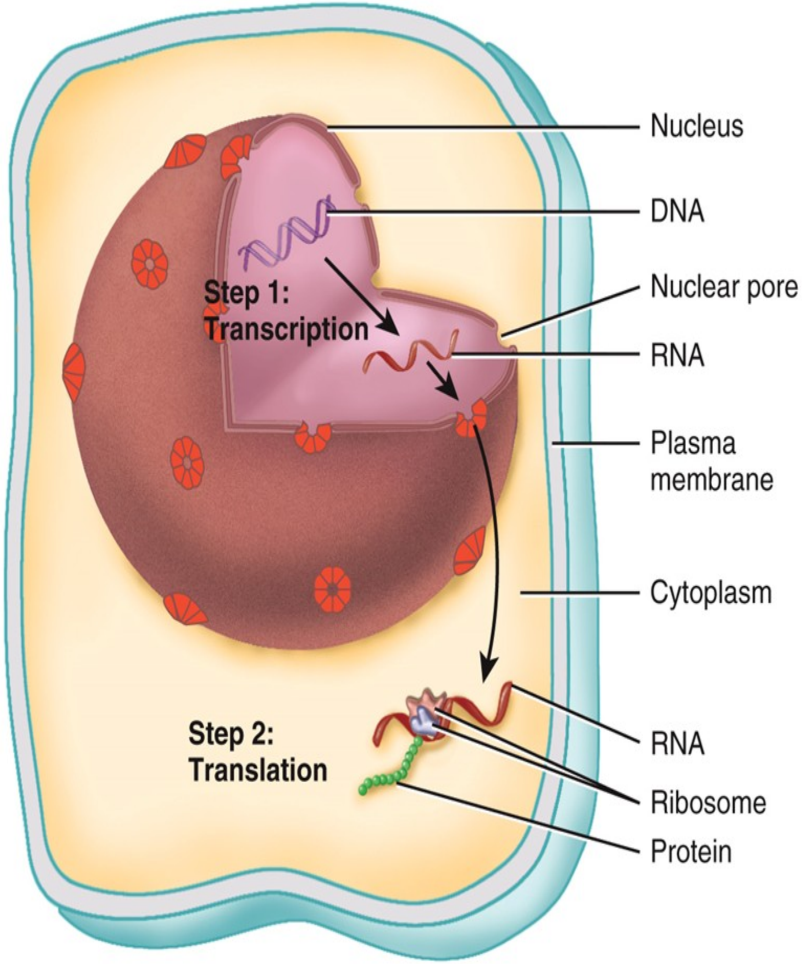
Gene Expression

- Transcription
  - Forms RNA from DNA
  - Occurs within the nucleus
- Translation
  - Creates a polypeptide from a sequence of mRNA
  - Occurs at ribosomes in cytoplasm



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Types of RNA

- Messenger RNA (mRNA)
  - Codes for the primary structure of a polypeptide
  - Codon – 3 nucleotide sequence that codes for one amino acid
- Ribosomal RNA (rRNA)
  - Along with proteins, forms ribosomes – site of polypeptide synthesis
- Transfer RNA (tRNA)
  - Brings amino acids to the ribosome
  - Anticodon – 3 nucleotide sequence that hydrogen bonds to codon sequence
    - Complementary to codon sequence

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Transcription

- Since RNA is single-stranded and DNA is double-stranded, only one strand of DNA serves as the template (referred to as the template strand)

- Only one gene gets transcribed at a time

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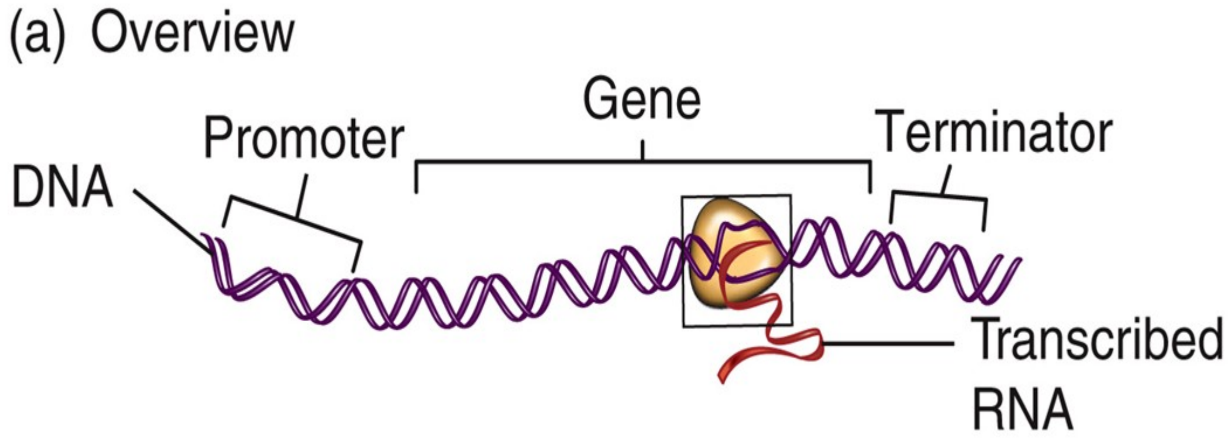
Transcription

- The promotor serves as an attachment site for transcription factors and RNA polymerase (the enzyme that links RNA nucleotides together)
- Transcription stops when a terminator DNA sequence is encountered



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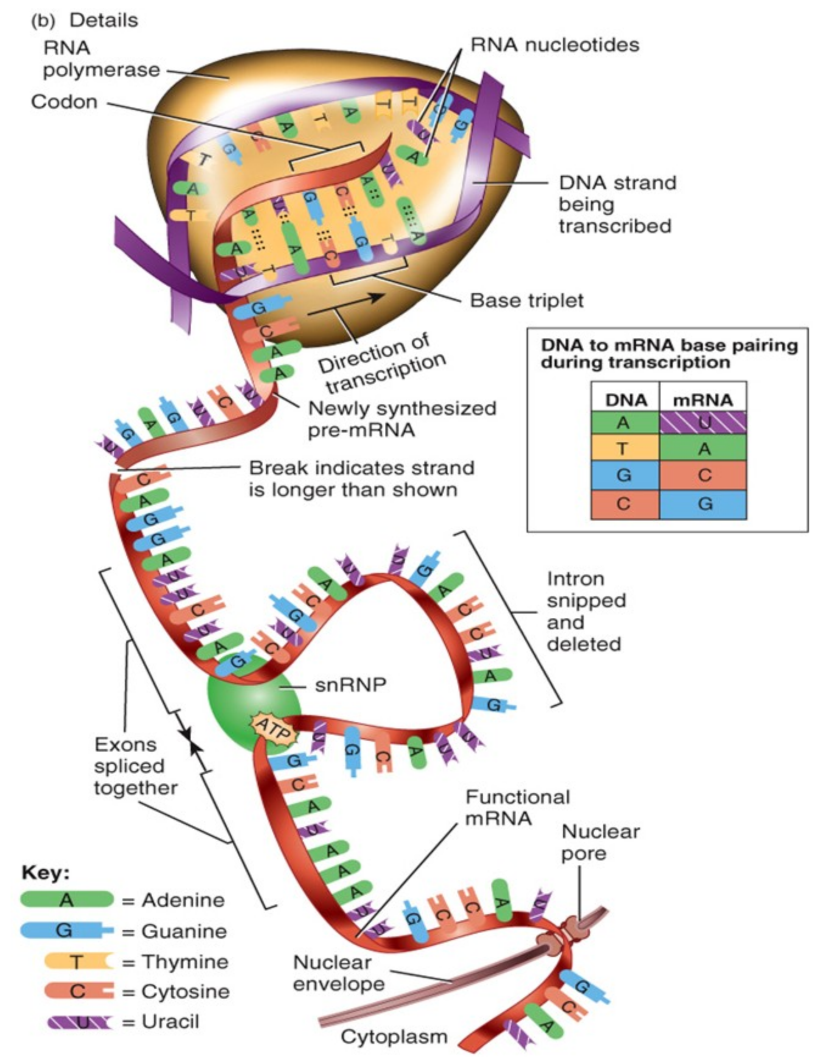
Transcription

- All RNA is transcribed in the same fashion
- mRNA has exons, which code for polypeptide, and introns, which need to be spliced out
    - Alternative splicing
        - Removal of exons as well as introns creates different mRNA molecules, which then result in different polypeptides



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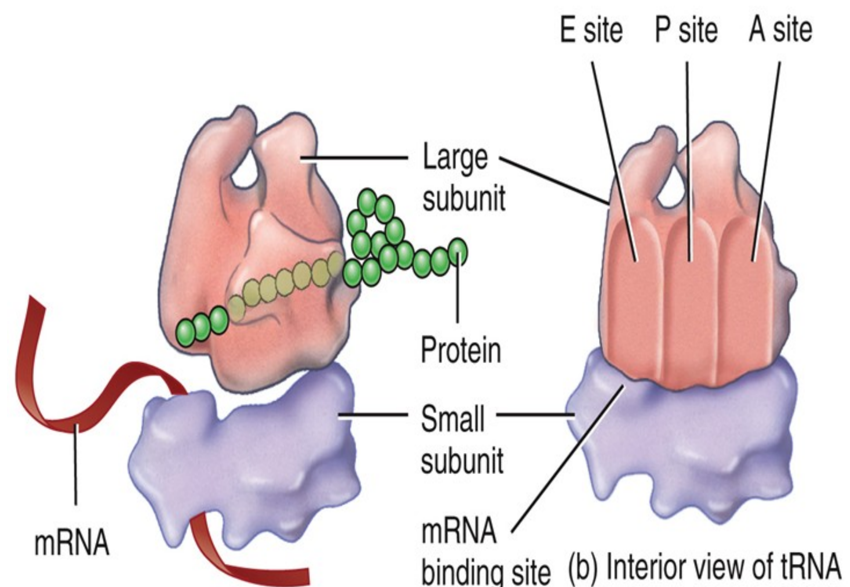
Translation

- Reads the mRNA nucleotide sequence to determine the amino acid sequence of a polypeptide

- Ribosomes have 3 sites that fit tRNA molecules
    - A site – tRNA first enters here
    - P site
    - E site – tRNA exits ribosome from here



(a) Components of a ribosome and their relationship to mRNA and protein during translation

(b) Interior view of tRNA binding sites

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Genetic code of mRNA

First base	Second base: U		C		A		G		Third base
U	UUU	Phe	UCU	Ser	UAU	Tyr	UGU	Cys	U
	UUC		UCC		UAC		UGC		C
	UUA	Leu	UCA		UAA	Stop	UGA	Stop	A
	UUG		UCG		UAG	Stop	UGG	Trp	G
C	CUU	Leu	CCU	Pro	CAU	His	CGU	Arg	U
	CUC		CCC		CAC		CGC		C
	CUA		CCA		CAA	Gln	CGA		A
	CUG		CCG		CAG		CGG		G
A	AUU	Ile	ACU	Thr	AAU	Asn	AGU	Ser	U
	AUC		ACC		AAC		AGC		C
	AUA		ACA		AAA	Lys	AGA	Arg	A
	AUG	Met	ACG		AAG		AGG		G
G	GUU	Val	GCU	Ala	GAU	Asp	GGU	Gly	U
	GUC		GCC		GAC		GGC		C
	GUA		GCA		GAA	Glu	GGA		A
	GUG		GCG		GAG		GGG		G

Figure 15-10
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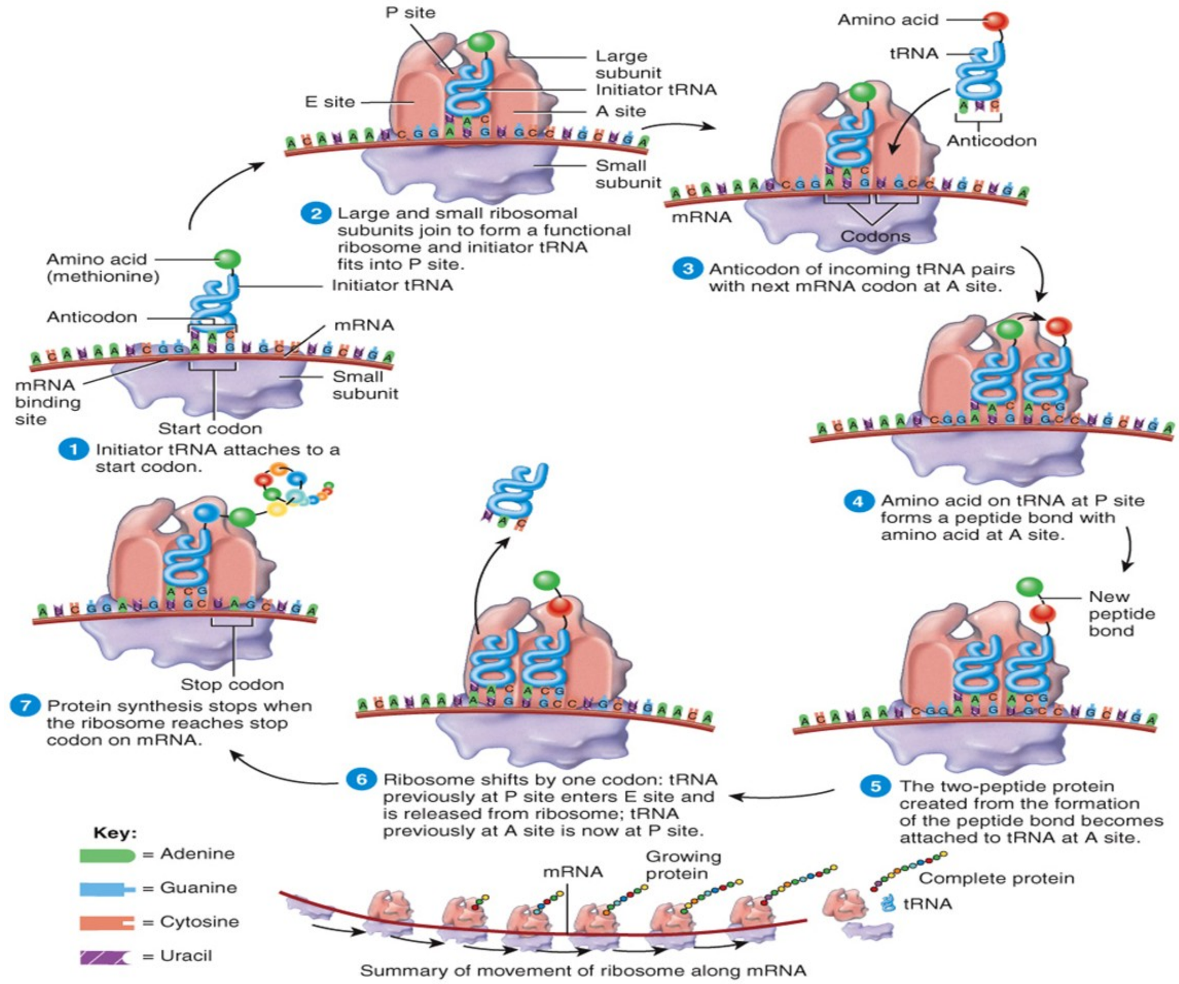
Translation



Protein Synthesis Interactions Animation:

[Protein Synthesis]

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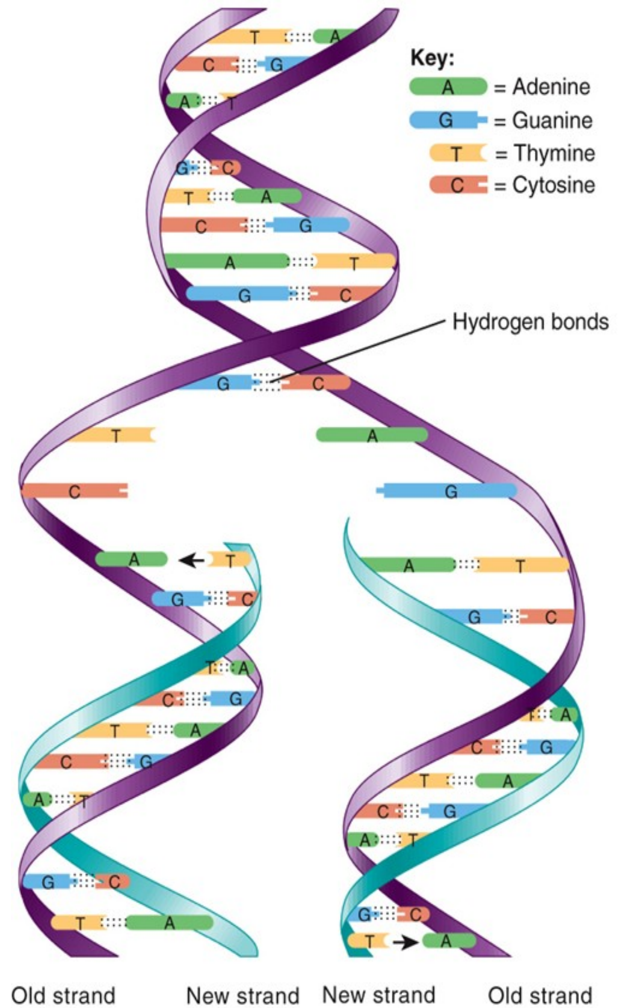
Replication of DNA

- Makes an identical copy of the original DNA molecule
  - Necessary before cell division
- Hydrogen bonds break, separating the two strands
- Each strand serves as a template for a new, second strand
  - Based on complementary base pairing



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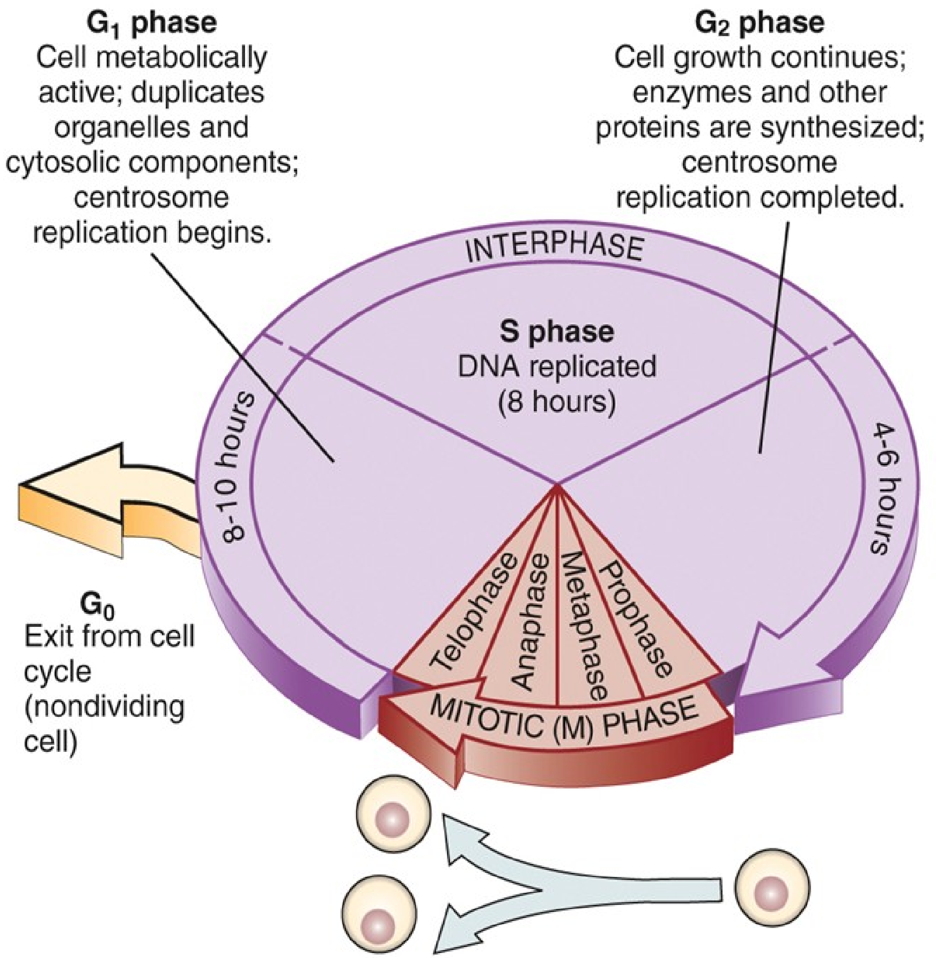
Cell Division – Mitosis

- Process in which cells reproduce themselves
- Cell cycle is divided into 3 main portions
    - Interphase
    - Mitosis
    - Cytokinesis
- The end result is two identical daughter cells



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Cell Cycle – Interphase

- G1
    - Cell is performing its day-to-day, minute-to-minute activities
    - Each chromosome consists of one DNA molecule/one chromatid
- S
    - DNA replicates
    - Each chromosome now consists of two identical DNA molecules/two sister chromatids
    - If cell is not going to divide, it will not enter the S stage
        - It will enter G0, which is similar to G1
- G2
    - Cell is further getting ready to divide

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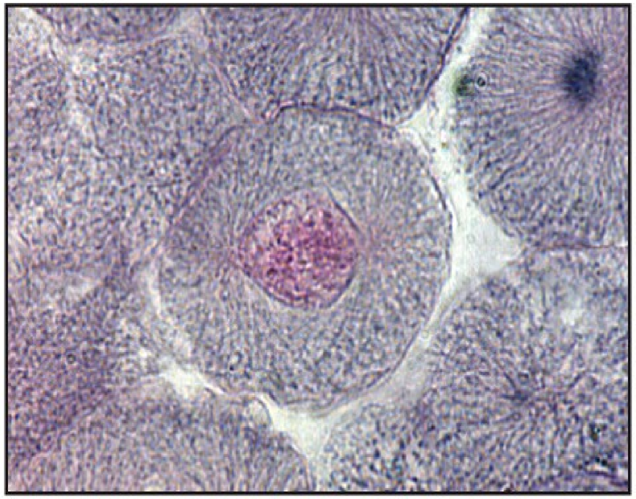
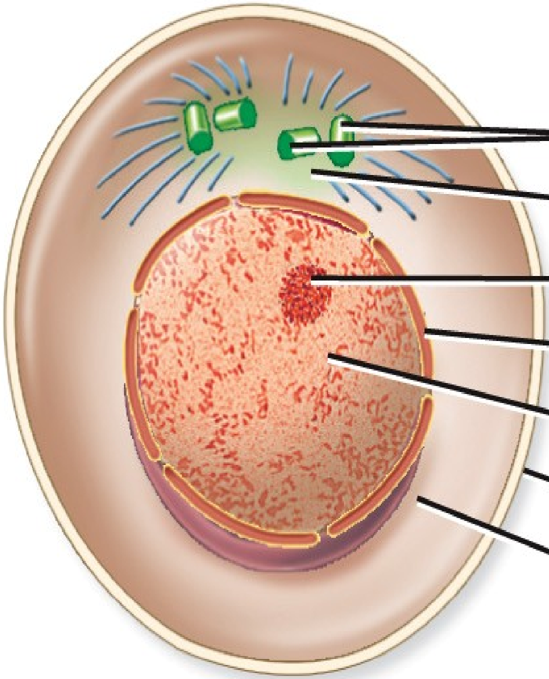
Interphase

Courtesy Michael Ross, University of Florida



LM all at 700x

(a) INTERPHASE



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Mitosis

- Mitosis occurs when the nucleus of a cell divides
- Mitosis results in the distribution of 2 sets of chromosomes into 2 separate nuclei
- Mitosis is divided into 4 steps:
  - Prophase
  - Metaphase
  - Anaphase
  - Telophase

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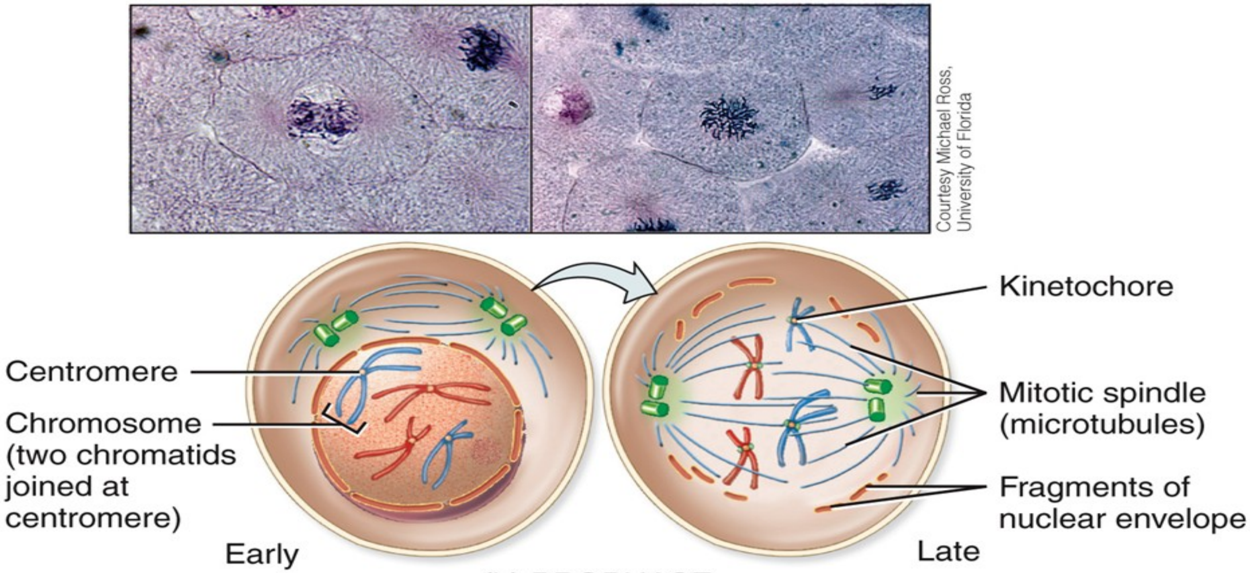
Mitosis: Prophase

- Chromatin condenses into chromosomes
- Centrosomes migrate to opposite poles of cell
- Microtubules/spindle apparatus forms
- Nuclear envelope breaks down



(b) PROPHASE

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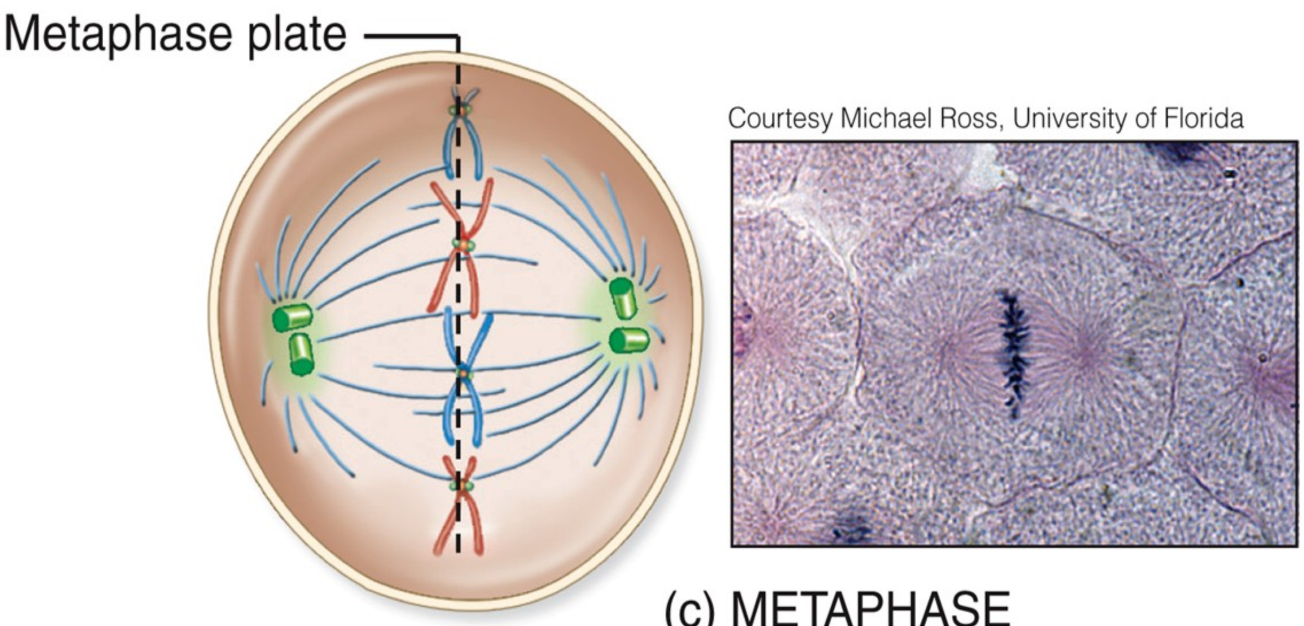
Mitosis: Metaphase

- Centromeres of chromosomes line up at the metaphase plate
    - Also known as equatorial plate



(c) METAPHASE

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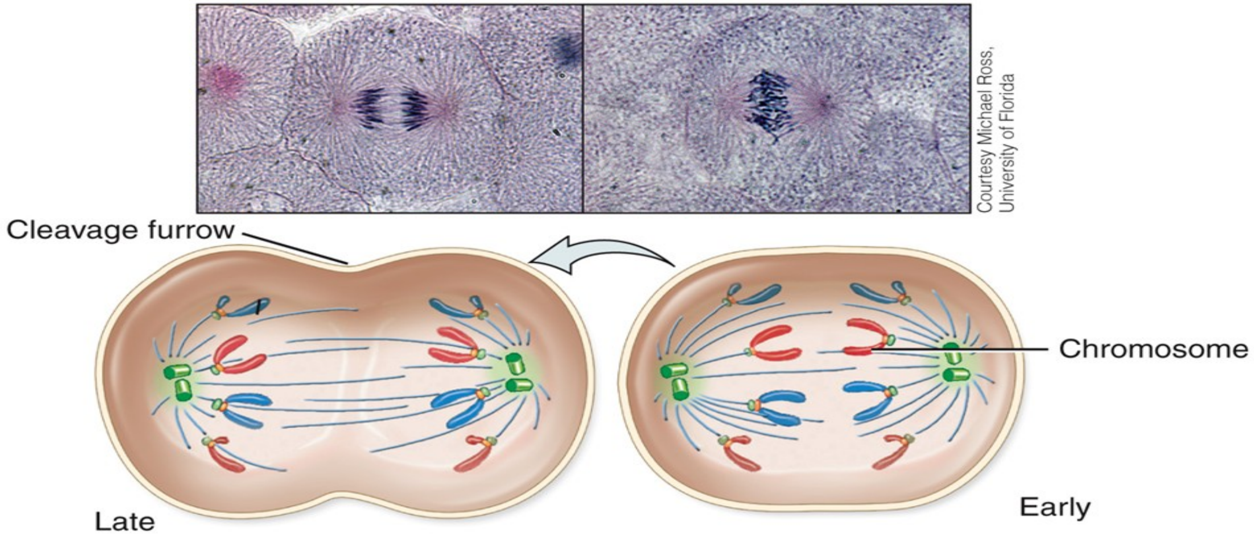
Mitosis: Anaphase

- Microtubules attached to centromeres shorten
    - This separates the sister chromatids
        - Each is now its own chromosome
        - Doubles the number of chromosomes in the cell



(d) ANAPHASE

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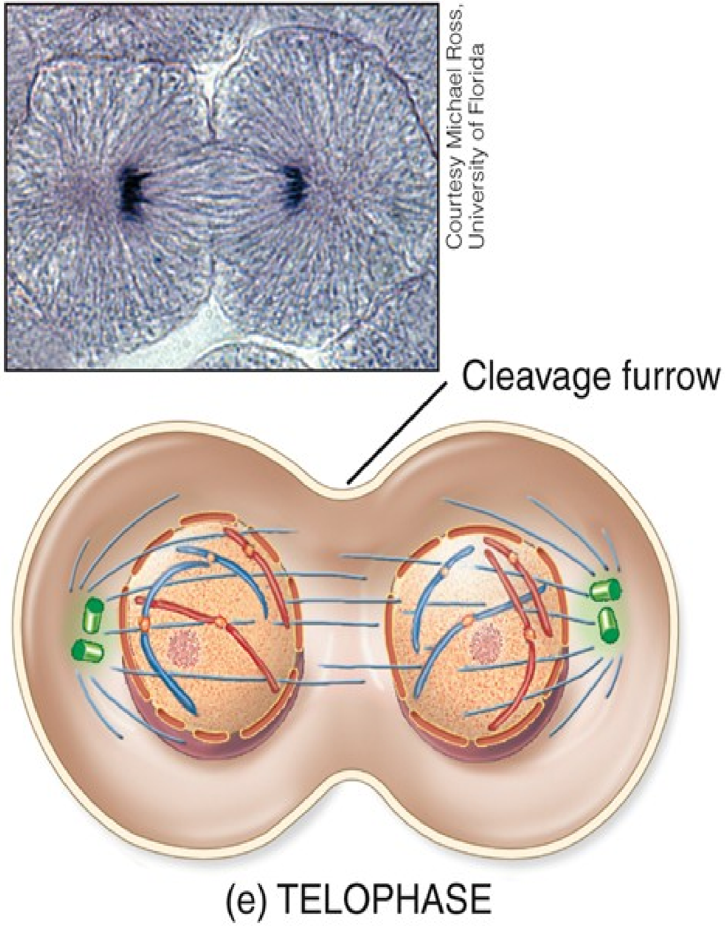
Mitosis: Telophase

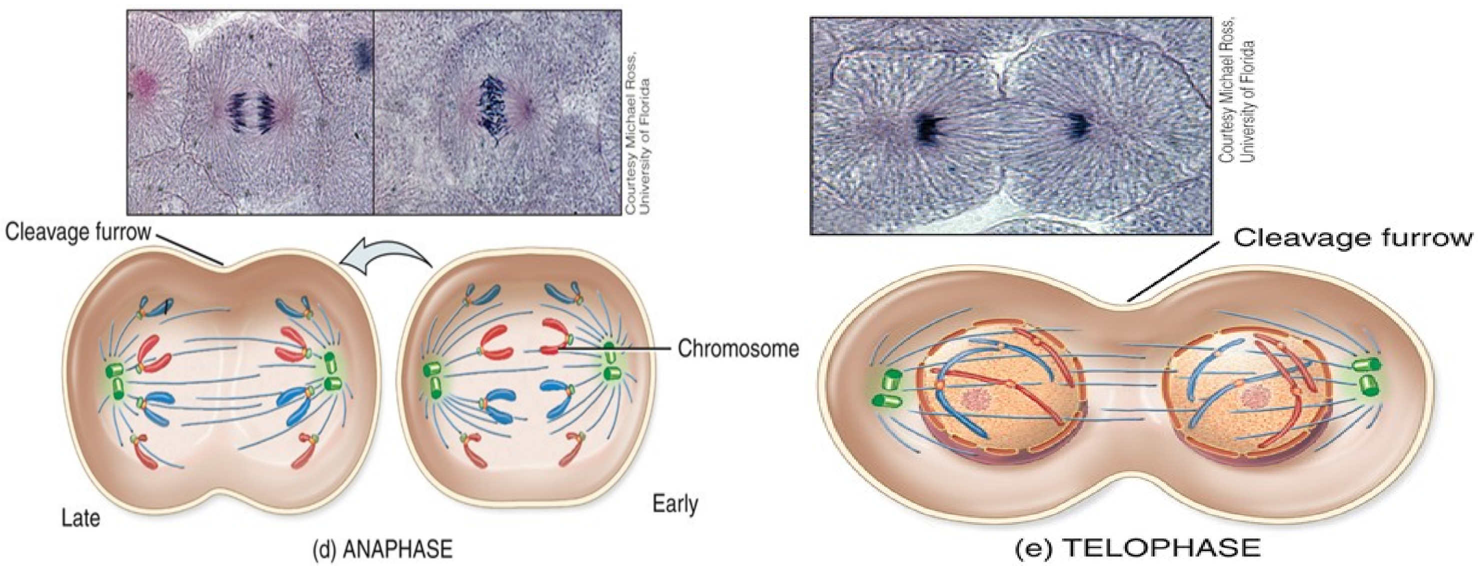
- Events are opposite to those of prophase
  - Nuclear membranes reform
  - Microtubules break down
  - Chromosomes decondense to more diffuse chromatin

- End result is one cell with 2 identical nuclei



Cytokinesis

- During cytokinesis a cleavage furrow forms and eventually the cytoplasm of the parent cell fully splits
  - Ring of actin/microfilaments at metaphase plate is the cause of this; deepens until ultimately cell is pinched in two



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Events of the Somatic Cell Cycle

Phase	Activity
Interphase	Period between cell divisions; chromosomes not visible under light microscope.
G_1 phase	Metabolically active cell duplicates most of its organelles and cytosolic components; replication of chromosomes begins. (Cells that remain in the G1 phase for a very long time, and possibly never divide again, are said to be in the G0 phase.)
S phase	Replication of DNA and centrosomes.
G_2 phase	Cell growth, enzyme and protein synthesis continue; replication of centrosomes complete.

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Events of the Somatic Cell Cycle

Phase	Activity
Mitotic phase	Parent cell produces identical cells with identical chromosomes; chromosomes visible under light microscope.
Mitosis	Nuclear division; distribution of two sets of chromosomes into separate nuclei.
Prophase	Chromatin fibers condense into paired chromatids; nucleolus and nuclear envelope disappear; each centrosome moves to an opposite pole of the cell.
Metaphase	Centromeres of chromatid pairs line up at metaphase plate.
Anaphase	Centromeres split; identical sets of chromosomes move to opposite poles of cell.
Telophase	Nuclear envelopes and nucleoli reappear; chromosomes resume chromatin form; mitotic spindle disappears.
Cytokinesis	Cytoplasmic division; contractile ring forms cleavage furrow around center of cell, dividing cytoplasm into separate and equal portions.

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Control of Cell Destiny

- Three possible destinies:
  - Remain alive and functioning without dividing
    - Some cell types only divide infrequently (ex: liver cells) and other cell types never divide (ex: adult neurons)
      - These cells would enter G0
  - Grow and divide
  - Die
    - Apoptosis – programmed cell death

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Cell Division – Meiosis

- Purpose of meiosis is to reduce the number of chromosomes by half in an egg or sperm
    - This ensures number of chromosomes remains the same with each generation

- Daughter cells formed via meiosis are genetically different from one another
    - Due to crossing over in Prophase I and random/independent assortment in Metaphase I

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Cell Division – Meiosis

- Starts with Interphase (identical to mitosis) followed by two rounds of division
  - Meiosis I separates homologous chromosomes
    - Homologous chromosomes are a pair – two of the same type
      - One was received from mother, the other from father
  - Meiosis II separates sister chromatids

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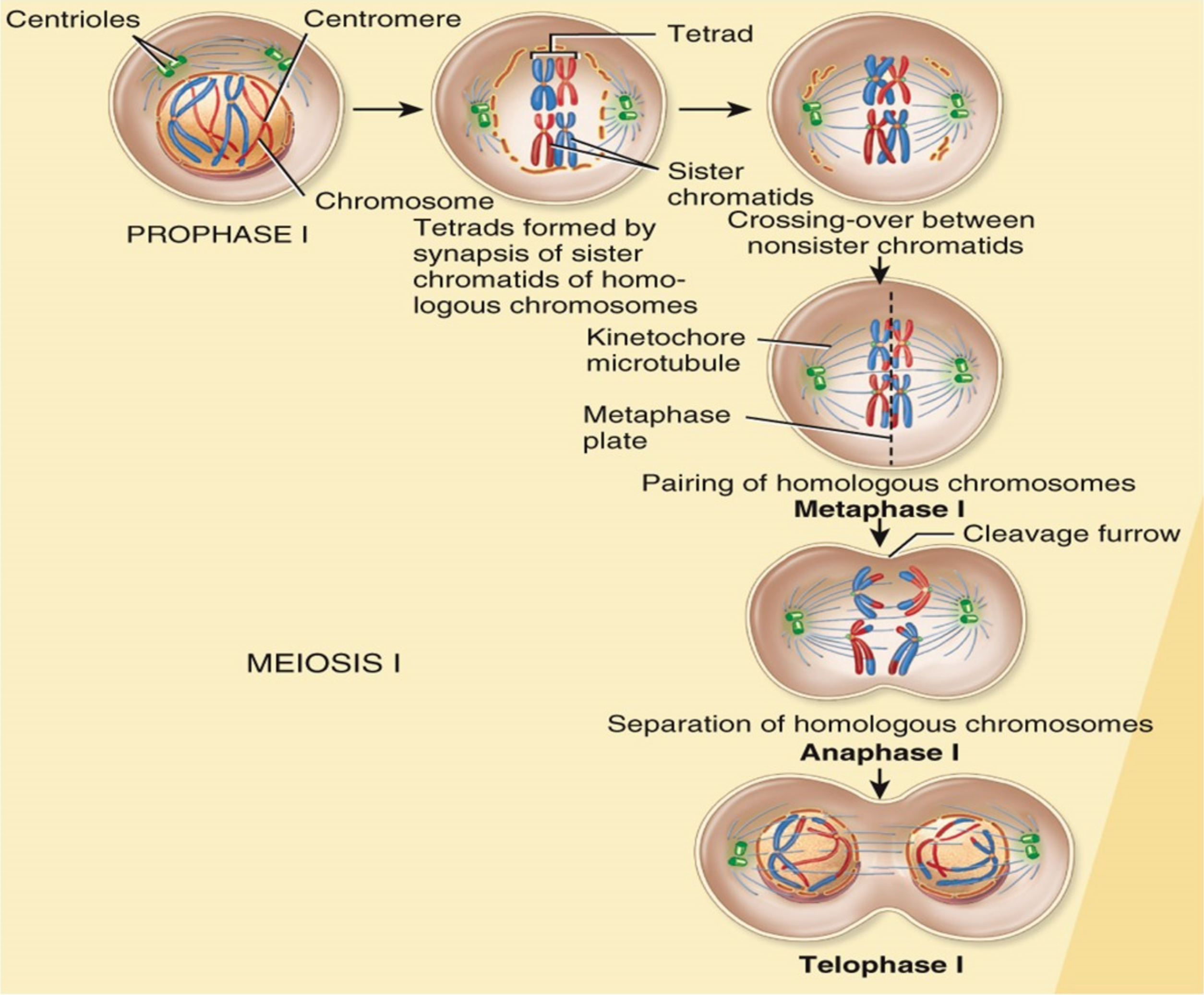
Reproductive Cell Division: Meiosis I



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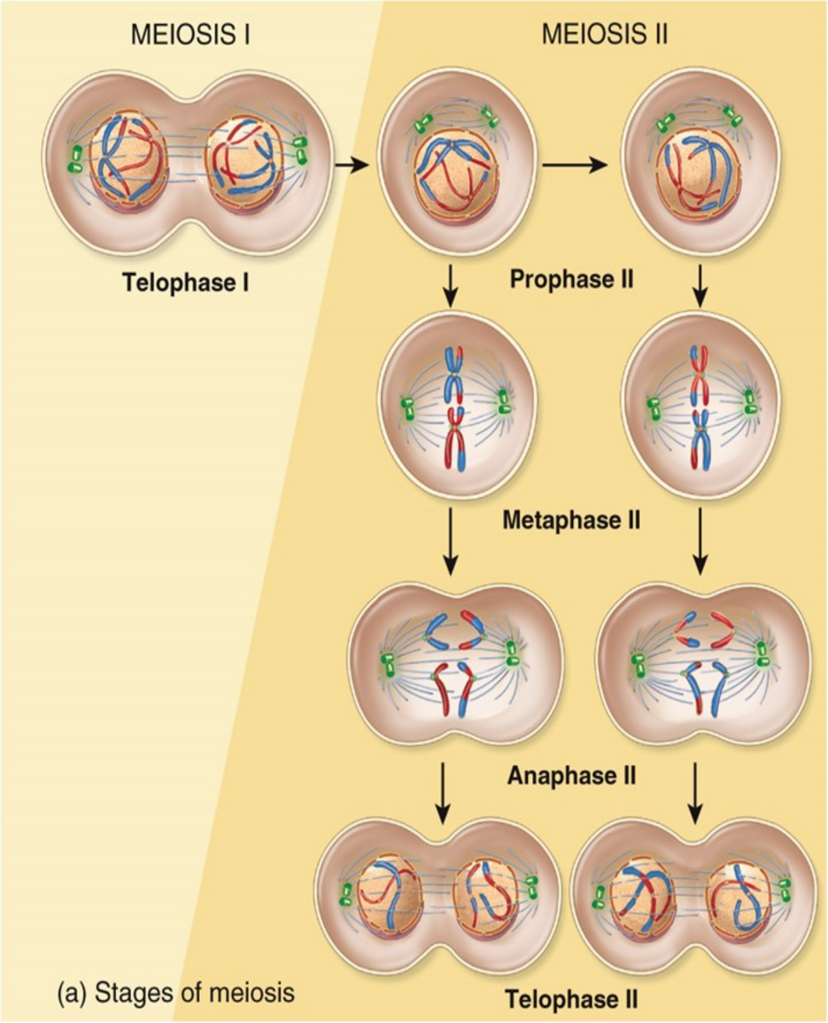
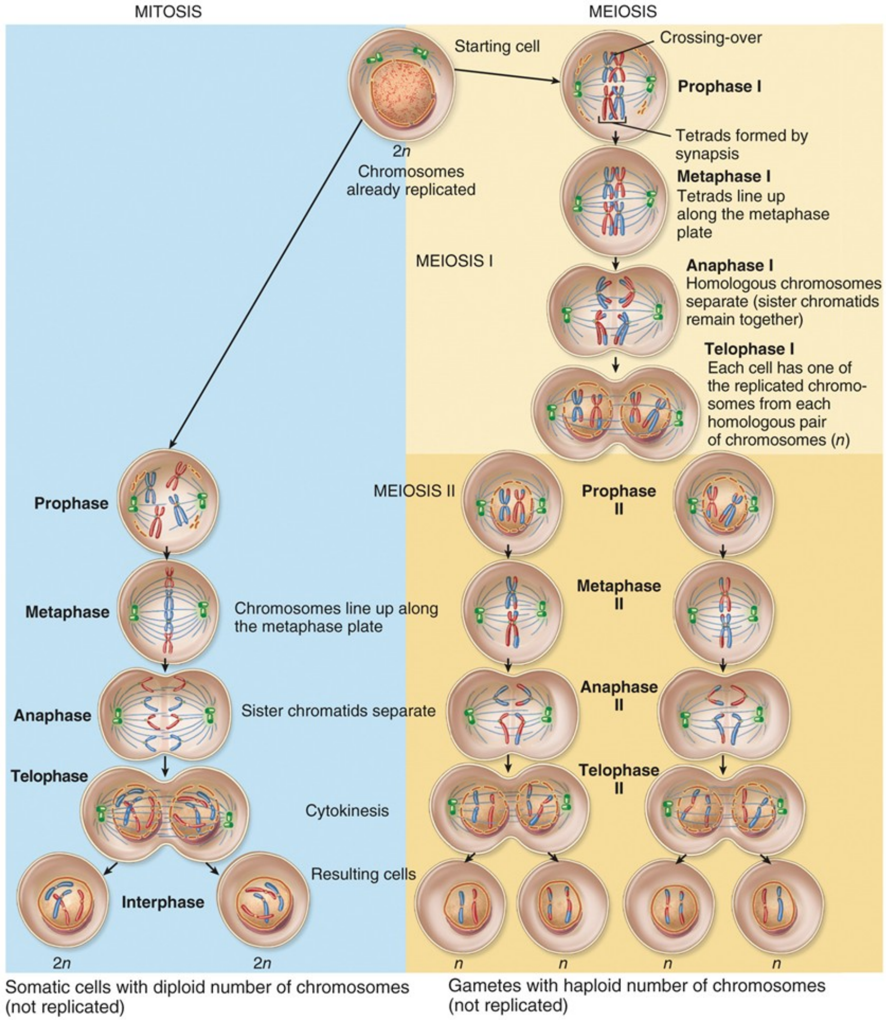
Reproductive Cell Division: Meiosis II



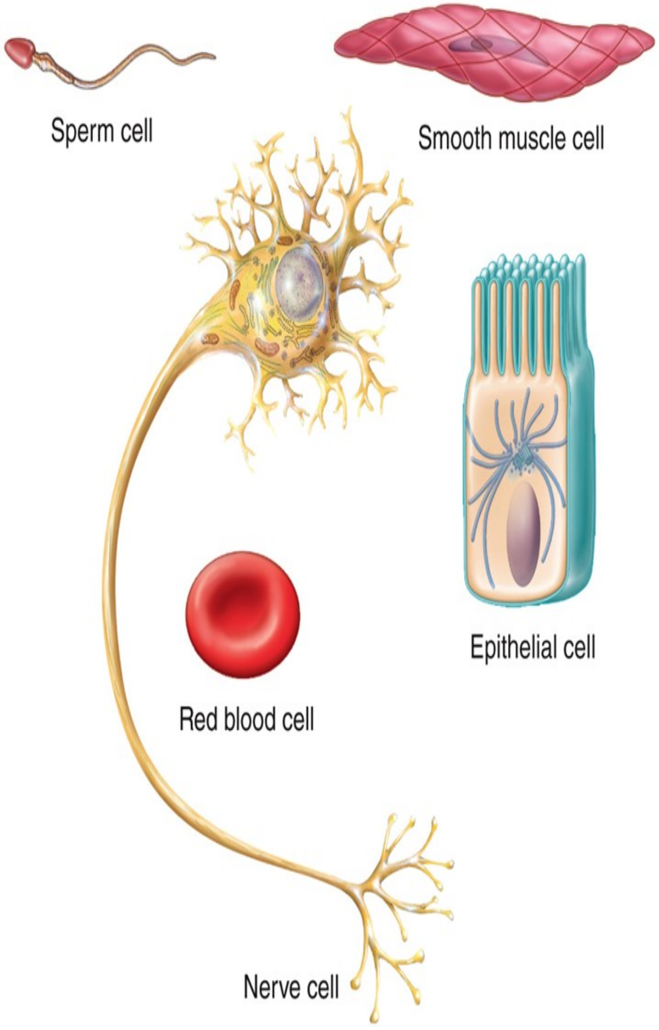
(a) Stages of meiosis

Comparison Between Mitosis (Left) and Meiosis (Right)



Cellular Diversity

- All somatic cells (any cell that isn't egg or sperm) have identical DNA to one another
  - The difference in appearance and function is the result of varying gene expression
    - Every cell has the same genes, but those genes are NOT all transcribed and therefore translated in every cell



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Aging and Cells

- As we age:
  - Our cells gradually deteriorate in their ability to function normally and in their ability to respond to environmental stresses
  - The numbers of our body cells decreases
  - We lose the integrity of the extracellular components of our tissues
  - Mutations and gene expression alterations accumulate in our DNA

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Tumors

- Unregulated cell growth
  - Due to cell dividing too often and/or not undergoing apoptosis (programmed cell death)
- Malignant
  - Has the ability to metastasize – spread to other parts of the body and establish a secondary tumor
  - Also known as cancer
    - Sarcoma – malignancy of connective tissue*
    - Carcinoma – malignancy of epithelial tissue*
- Benign
  - Cannot metastasize

*tissue types will be discussed in chapter 4

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Disorders: Cancer

- Types of Cancer
  - Melanoma
  - Sarcoma
  - Osteogenic sarcoma
  - Leukemia
  - Lymphoma

- Growth and spread
  - Angiogenesis – formation of blood vessels

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Disorders: Cancer

- Causes
  - Carcinogens
  - Oncogenes
  - Oncogenic viruses
- Treatments
  - Surgery
  - Chemotherapy
  - Radiation
  - Others: immunotherapy, targeted therapy, stem cell transplants, hormone therapy, CAM

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